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Brand Names

Ambien, Ambien CR, Edluar, Intermezzo, Zolpimist

Indication Specific Dosing

For the treatment of insomnia

For the short-term treatment of insomnia characterized by difficulty with sleep initiation

Oral dosage (e.g., immediate-release oral tablets, such as Ambien)

Adult Females

5 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening; this is also the recommended dose for debilitated adults. Administer as a single dose and do not re-administer during the same night. Use the lowest effective dose and shortest duration possible. Max: 10 mg/day, but only if needed; the higher dose is more likely to cause next-day impairment. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Adult Males

5 to 10 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening; 5 mg PO immediately before bedtime is recommended for debilitated adults. Administer as a single dose and do not re-administer during the same night. Many men respond with sufficient efficacy at 5 mg PO at bedtime. Use the lowest effective dose and shortest duration possible. Max: 10 mg/day, but only if needed; the higher dose is more likely to cause next-day impairment.

Geriatric Adults

5 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Administer as a single dose and do not

re-administer during the same night. Use the lowest effective dose for the shortest treatment duration possible. Geriatric and debilitated adults are particularly sensitive to the effects of zolpidem. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Children† and Adolescents†

Efficacy and safety have not been established. Zolpidem did not significantly decrease sleep latency vs. placebo in a controlled trial of 201 pediatric patients (6 to 17 years) with ADHD-related insomnia given zolpidem 0.25 mg/kg/day (Max: 10 mg) PO 30 minutes prior to bedtime. Psychiatric and CNS effects were more frequent in the zolpidem group vs. the placebo group. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Sublingual dosage (sublingual tablets; e.g., Edluar)

Adult Females

5 mg sublingually immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Use the lowest effective dose for the shortest duration possible. Administer as a single dose and do not re-administer during the same night. Max: 10 mg/day if needed. Debilitated adult Max: 5 mg/day. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Adult Males

5 to 10 mg sublingually immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Efficacy is sufficient for many men at 5 mg PO at bedtime. Administer as a single dose and do not re-administer during the same night. Max: 10 mg/day if needed. Debilitated adult Max: 5 mg/day. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Geriatric Adults

5 mg sublingually immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Use the lowest effective dose for the shortest duration possible. Administer as a single dose and do not re-administer during the same night. Geriatric patients may be particularly sensitive

to the effects of the drug. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Oral dosage (e.g., immediate-release oral capsules)

Adults females

For patients with an inadequate response to zolpidem 5 mg/night, initiate 7.5 mg capsules PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Administer as a single dose and do not re-administer during the same night.Patients requiring a lower or higher dose of zolpidem should receive tablets, as the capsule formulation is only available in 7.5 mg strength. Max: 7.5 mg/day.

Adult males

For patients with an inadequate response to zolpidem 5 mg/night or needing a lower dose than 10 mg/night, initiate 7.5 mg capsules PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Administer as a single dose and do not re-administer during the same night. Patients requiring a lower or higher dose of zolpidem should receive tablets, as the capsule formulation is only available in 7.5 mg strength. Max: 7.5 mg/day if needed. Administer as a single dose and do not re-administer during the same night. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

For insomnia characterized by difficulties with sleep onset and/or sleep maintenance

Oral dosage (extended-release tablets; e.g., Ambien CR)

Adult Females

6.25 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Use the lowest effective dose for shortest duration possible. Administer as a single dose and do not re-administer during the same night. Max: 12.5 mg/day if needed; however, the higher dose is more likely to cause next-day impairment. Debilitated adult Max: 6.25 mg/night. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Adult Males

6.25 to 12.5 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening. Many men have sufficient efficacy at 6.25 mg PO before bedtime. Use the lowest effective dose for shortest duration possible. Administer as a single dose and do not re-administer during the same night. Max: 12.5 mg/day if needed; however, the higher dose is more likely to cause next-day impairment. Debilitated adult Max: 6.25 mg/day. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Geriatric Adults

6.25 mg PO immediately before bedtime and with at least 7 to 8 hours remaining before the planned time of awakening is the recommended and max dose. Administer as a single dose and do not re-administer during the same night. Geriatric and debilitated adults may be particularly sensitive to the effects of the drug. Use the lowest effective dose for shortest duration possible. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

For middle-of-the night awakening that is followed by difficulty returning to sleep

Sublingual dosage (sublingual tablets; Intermezzo ONLY)

Adult Females

1.75 mg sublingually taken once per night if needed for a middle-of-the night awakening followed by difficulty returning to sleep is the recommended and max dose. Use the lowest effective dose for shortest duration possible. Use only if the patient has at least 4 hours of bedtime remaining before the planned time of waking. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Adult Males

3.5 mg sublingually taken once per night if needed for a middle-of-the night awakening followed by difficulty returning to sleep. Use a lower dose of 1.75 mg/night for men taking other CNS depressants. Use only if the patient has at least 4 hours of bedtime remaining before the planned time of waking. Use the lowest effective dose for shortest duration possible. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Geriatric Adults over 65 years of age

1.75 mg sublingually taken once per night for a middle-of-the night awakening followed by difficulty returning to sleep. Use the lowest effective dose for shortest duration possible. Use only when there are at least 4 hours of bedtime remaining before the planned time of waking. Coadministration of certain drugs may need to be avoided or dosage adjustments may be necessary; review drug interactions.

Contraindications And Precaution

Drug Interactions

The coadministration of certain medications may lead to harm and require avoidance or therapy modification; review all drug interactions prior to concomitant use of other medications.

Hypersensitivity

This medication is contraindicated in patients with a history of hypersensitivity to it or any of its components. While zolpidem belongs to the nonbenzodiazepine, benzodiazepine receptor agonist (NBRA) class, the chemical structures of the agents in this class are distinct. Cross-sensitivity to one NBRA does not imply cross-sensitivity will occur to another agent in the class.

activities requiring coordination and concentration, driving or operating machinery, drug-induced complex sleep-related behaviors, females

Zolpidem and other non-benzodiazepine receptor agonists (NBRAs) are contraindicated in people with a history of drug-induced complex sleep-related behaviors. Sedative-hypnotics, especially NBRAs like zolpidem, can cause complex sleep-related behaviors (e.g., sleep-driving, eating, sexual activity, or making phone calls) with no memory of these events. These effects can occur with or without concomitant use of alcohol or other CNS depressants. Rarely, serious injuries or death have occurred. Patients should discontinue zolpidem immediately and contact their healthcare provider if such episodes occur. Care teams and patients are encouraged to report adverse events to the FDA MedWatch Safety Information and Adverse Event Reporting Program. Zolpidem has rapid onset and CNS depressant effects, and should be taken immediately before bed, with at least 7 to 8 hours remaining for sleep when used for insomnia or 4 hours for treatment of middle-of-the-night awakenings. Patients should avoid driving or operating machinery, or engaging in activities requiring coordination and concentration for at least

8 hours after immediate-release formulations and impairment may last up to 24 hours following ingestion of an extended-release formulation. The risk for next-day impairment is increased if zolpidem is taken without adherence to recommended hours for sleep or at doses higher than recommended. Anterograde amnesia may occur and can be particularly evident at zolpidem doses above 10 mg/day. Use with alcohol or other CNS depressants increases the risk of impairment and adverse effects, and patients should avoid alcoholic beverages. Consider lower doses of zolpidem when other CNS depressants are prescribed concurrently. Due to gender differences, lower initial zolpidem doses are recommended for females, as they may have higher plasma levels the next morning, increasing impairment risk.

respiratory insufficiency

Sedative-hypnotics have the potential to suppress respiratory drive, and caution is advised when using zolpidem in people with compromised respiratory function (respiratory insufficiency). Reports of respiratory insufficiency in people receiving zolpidem, many of whom had pre-existing respiratory impairment, have occurred. The risk of respiratory depression should be considered before using zolpidem in people with respiratory insufficiency, including severe chronic obstructive pulmonary disease (COPD), sleep apnea, myasthenia gravis, or individuals with concomitant opioid use.

substance abuse disorder

Zolpidem should be used with caution in people with a history of a substance abuse disorder. People with a history of addiction or abuse of drugs or alcohol are at increased risk for misuse, abuse, or addiction and should be monitored carefully when receiving zolpidem or any hypnotic. Additionally, people taking zolpidem are at risk of physical and/or psychological dependence and this risk increases with dose and duration of treatment. People with dependence on zolpidem may experience withdrawal signs and symptoms following abrupt discontinuation. To minimize these risks, zolpidem should be used at the lowest effective dose for the shortest duration of treatment possible, and reassess the patient periodically before continuing treatment for an extended period of time.

depression, suicidal ideation

Worsening of depression and suicidal ideation, including completed suicides, have been reported during the use of hypnotics, primarily in people with pre-existing depression. Immediately evaluate patients with worsening depression, emergent suicidal ideation (suicidal thoughts or actions), or other new or worsening adverse behaviors during treatment with zolpidem. A variety of abnormal thinking and behavioral changes have

been reported to occur in association with the use of sedative/hypnotics, including zolpidem. Some of these changes included decreased inhibition (e.g., aggressiveness and extroversion that seemed out of character), bizarre behavior, agitation and depersonalization. Visual and auditory hallucinations have been reported. Amnesia and other neuropsychiatric symptoms may occur unpredictably. Some adverse effects of zolpidem appear to be dose-related; use of the lowest effective dose may mitigate these effects. Because intentional overdose is more common in people with pre-existing depression, zolpidem should be prescribed in the smallest quantity consistent with good patient management to reduce the risk of overdose.

Child-Pugh class A, Child-Pugh class B, Child-Pugh class C, hepatic failure

Zolpidem dosage adjustments are recommended in people with mild (Child-Pugh class A) or moderate (Child-Pugh class B) hepatic impairment due to reduced clearance and subsequently increased systemic exposure. Use is not recommended in severe hepatic impairment (Child-Pugh class C) or hepatic failure; increased zolpidem exposure in this population may also increase the risk of precipitating hepatic encephalopathy.

geriatric

Geriatric adults are more sensitive to the effects of zolpidem, necessitating lower initial doses and close monitoring. Since zolpidem can cause drowsiness and impairs consciousness, there is an increased risk of falls and severe injuries in older adults. Specifically, doses exceeding 5 mg/day of immediate-release zolpidem or 6.25 mg/day of extended-release zolpidem may heighten the risk of falls and mental status changes. Discontinuing zolpidem, especially at higher doses, can lead to rebound insomnia; however, at recommended doses, no objective evidence of rebound insomnia has been observed. During studies, older adults reported impaired sleep on the first night after stopping zolpidem doses above 5 mg/day. The Beers Criteria state that zolpidem and other nonbenzodiazepine, benzodiazepine-receptor agonists (NBRAs), are potentially inappropriate for use in geriatric adults. NBRAs may cause adverse effects similar to benzodiazepines, including falls, fractures, and delirium, which are particularly concerning in older adults. Their use is associated with increased emergency department visits, hospitalizations, and motor vehicle crashes, with minimal benefit in improving sleep latency and duration. Avoid NBRA use in individuals with dementia or cognitive impairment, or those at high risk of delirium. Additionally, avoid NBRAs in geriatric adults with a history of falls or fractures unless no safer alternatives are available. If an NBRA must be prescribed, consider reducing other CNS-active medications and employing strategies to mitigate fall risk. The U.S. Omnibus Budget Reconciliation Act (OBRA) regulates the use of sedative/hypnotic agents for residents of

long-term care facilities (LTCFs); when a drug is being used to induce sleep or treat a sleep disorder, the facility should attempt periodic tapering of the medication as stated in the OBRA guidelines or provide documentation of continued medical necessity.

pregnancy

The risk of zolpidem use during pregnancy should be carefully weighed against the risk of untreated sleep disorders in the pregnant person. Available data from observational studies, birth registries, and case reports on the use of zolpidem during human pregnancy have not reported a clear association with the drug and major birth defects; however, hypnotic benzodiazepine receptor agonist (HBRA) use in early pregnancy may be associated with a risk for nonteratogenic effects. The HBRAs, including zolpidem, have been shown to cross the human placenta. In a study examining data from the Swedish Medical Birth Registry, the birth outcomes for 603 infants who were exposed to zolpidem in early pregnancy were compared to nonexposed infants. In this study, the use of zolpidem was not associated with an increased risk for congenital malformations. There were 4 cases of intestinal malformations reported in infants exposed to HBRAs; 2 of these cases were in infants exposed to zolpidem. The authors noted the presence of several potential confounders, such as exposure to other medications, smoking status, and maternal age, that may have contributed to these malformations. Published studies of other hypnotic benzodiazepine receptor agonists (HBRAs) like zolpidem do not indicate an increased risk of congenital malformations at typical doses. However, one study has reported neural tube defects in an infant following high-dose exposure to zolpidem in the first trimester of pregnancy. In a systematic review and meta-analysis developed to assess for risks associated with use of HBRAs in pregnancy, the authors noted that pregnancy exposure to HBRAs was not associated with an increased risk of congenital malformations but was associated with increased risks of preterm birth, low birth weight, and being small for gestational age compared to unexposed infants. Similarly, a population-based cohort study determined that HBRA use in early pregnancy may be associated with an increased risk of infants being born small for gestational age, even after controlling for numerous confounding variables. Additional studies are needed to determine the true risk and incidence of these effects.

breast-feeding

Use zolpidem with caution during breast-feeding. Zolpidem is excreted into breast milk, and there are reports of excess sedation in infants exposed to zolpidem through breast milk. Monitor the exposed infants for excess sedation, hypotonia, poor weight gain, and respiratory depression. To minimize infant drug exposure, it has been suggested that patients interrupt breast-feeding, and pump and discard breast milk during treatment and for 23 hours (about 5 half-lives) after zolpidem administration. Expert opinion is

divided, with some experts recommending against zolpidem use while breast-feeding, while others suggest zolpidem could be used safely in older breast-fed infants that do not breast-feed overnight after the maternal dose. In one small study of breast-feeding people (n = 5) who received a single 20 mg dose of zolpidem on postpartum day 3 or 4, the mean milk to maternal plasma concentration ratio was 0.13 (range, 0.11 to 0.18) three hours after drug administration. The amount of zolpidem measured in the breast milk samples represented 0.004% to 0.019% of the maternal administered dose. No detectable zolpidem was measured in subsequent milk samples taken 13 and 16 hours after the dose. Breast-feeding was discontinued for 24 hours after drug administration; therefore, infant exposure was not assessed. The transfer of zaleplon, an alternative sedative, into breast milk has also been assessed; however, the effects of zaleplon exposure on the breast-feeding infant have not been formally evaluated.

neonates and infants exposed to this medication in utero

Monitor exposed neonates and infants exposed to this medication in utero during the third trimester for signs and symptoms of respiratory depression and sedation, and manage accordingly. Withdrawal symptoms may also be possible during the postnatal period. A limited number of exposed neonatal cases of moderate to severe respiratory depression requiring aspiration, artificial ventilation, or intratracheal intubation have been reported. The majority of neonates recovered within hours to a few weeks after birth once treated. Neonatal flaccidity has also been reported in infants born to individuals who have received sedatives during pregnancy.

Pregnancy And Lactation

The risk of zolpidem use during pregnancy should be carefully weighed against the risk of untreated sleep disorders in the pregnant person. Available data from observational studies, birth registries, and case reports on the use of zolpidem during human pregnancy have not reported a clear association with the drug and major birth defects; however, hypnotic benzodiazepine receptor agonist (HBRA) use in early pregnancy may be associated with a risk for nonteratogenic effects. The HBRAs, including zolpidem, have been shown to cross the human placenta. In a study examining data from the Swedish Medical Birth Registry, the birth outcomes for 603 infants who were exposed to zolpidem in early pregnancy were compared to nonexposed infants. In this study, the use of zolpidem was not associated with an increased risk for congenital malformations. There were 4 cases of intestinal malformations reported in infants exposed to HBRAs; 2 of these cases were in infants exposed to zolpidem. The authors noted the presence of several potential confounders, such as exposure to other medications, smoking status, and maternal age, that may have contributed to these malformations. Published studies

of other hypnotic benzodiazepine receptor agonists (HBRAs) like zolpidem do not indicate an increased risk of congenital malformations at typical doses. However, one study has reported neural tube defects in an infant following high-dose exposure to zolpidem in the first trimester of pregnancy. In a systematic review and meta-analysis developed to assess for risks associated with use of HBRAs in pregnancy, the authors noted that pregnancy exposure to HBRAs was not associated with an increased risk of congenital malformations but was associated with increased risks of preterm birth, low birth weight, and being small for gestational age compared to unexposed infants. Similarly, a population-based cohort study determined that HBRA use in early pregnancy may be associated with an increased risk of infants being born small for gestational age, even after controlling for numerous confounding variables. Additional studies are needed to determine the true risk and incidence of these effects.

Interactions

Acetaminophen; Aspirin, ASA; Caffeine: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Acetaminophen; Aspirin; diphenhydrAMINE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Caffeine: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Acetaminophen; Caffeine; Dihydrocodeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative

treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Acetaminophen; Caffeine; Pyrilamine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Acetaminophen; Chlorpheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Chlorpheniramine; Dextromethorphan: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Chlorpheniramine; Dextromethorphan; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Chlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate)

Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Chlorpheniramine; Phenylephrine : (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Acetaminophen; Dextromethorphan; Doxylamine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; diphenhydrAMINE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Acetaminophen; HYDROcodone: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Acetaminophen; oxyCODONE: (Major) Concomitant use of oxycodone with zolpidem

may lead to additive respiratory and/or CNS depression. Hypotension, profound sedation, coma, respiratory depression, or death may occur. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. If zolpidem is used concurrently with oxycodone, a reduced dosage of oxycodone and/or zolpidem is recommended; use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor for sedation and respiratory depression.
Acetaminophen; Pamabrom; Pyrilamine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Adagrasib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with adagrasib is necessary. Concurrent use may increase zolpidem exposure. Zolpidem is a CYP3A substrate and adagrasib is a strong CYP3A inhibitor. Coadministration with other strong CYP3A inhibitors increased the AUC of zolpidem by 34% to 70%.
ALFentanil: (Moderate) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
ALPRAZolam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Amitriptyline: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.

Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Amobarbital: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Amoxapine: (Moderate) CNS depressants should be combined cautiously with amoxapine because they could cause additive depressant effects and possible respiratory depression or hypotension.
Amoxicillin; Clarithromycin; Omeprazole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with clarithromycin is necessary. Zolpidem is a CYP3A4 substrate and clarithromycin is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Apalutamide: (Major) Coadministration of zolpidem with apalutamide is not recommended due to decreased plasma concentrations of zolpidem. Zolpidem is a CYP3A4 substrate and apalutamide is a strong CYP3A4 inducer. Coadministration with another strong CYP3A4 inducer decreased zolpidem exposure by 73% and significantly reduced the pharmacodynamic effects of zolpidem.
Apomorphine: (Moderate) Apomorphine causes significant somnolence. Concomitant administration of apomorphine and CNS depressants could result in additive depressant effects. A reduction in the dose of one or both drugs should be considered to minimize additive sedative effects. In addition, the risk of next-day psychomotor impairment is increased during co-administration of eszopiclone and other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Apraclonidine: (Minor) No specific drug interactions were identified with systemic agents and apraclonidine during clinical trials. Theoretically, apraclonidine might potentiate the effects of CNS depressant drugs such as the anxiolytics, sedatives, and hypnotics, including barbiturates or benzodiazepines.
Aprepitant, Fosaprepitant: (Moderate) Use caution if zolpidem and a multi-day regimen of oral aprepitant are used concurrently and monitor for an increase in zolpidem-related adverse effects for several days after administration. Zolpidem is a CYP3A4 substrate.

Aprepitant, when administered as a 3-day oral regimen (125 mg/80 mg/80 mg), is a moderate CYP3A4 inhibitor and inducer and may increase plasma concentrations of zolpidem. For example, a 5-day oral aprepitant regimen increased the AUC of another CYP3A4 substrate, midazolam (single dose), by 2.3-fold on day 1 and by 3.3-fold on day 5. After a 3-day oral aprepitant regimen, the AUC of midazolam (given on days 1, 4, 8, and 15) increased by 25% on day 4, and then decreased by 19% and 4% on days 8 and 15, respectively. As a single 125 mg or 40 mg oral dose, the inhibitory effect of aprepitant on CYP3A4 is weak, with the AUC of midazolam increased by 1.5-fold and 1.2-fold, respectively. After administration, fosaprepitant is rapidly converted to aprepitant and shares many of the same drug interactions. However, as a single 150 mg intravenous dose, fosaprepitant only weakly inhibits CYP3A4 for a duration of 2 days; there is no evidence of CYP3A4 induction. Fosaprepitant 150 mg IV as a single dose increased the AUC of midazolam (given on days 1 and 4) by approximately 1.8-fold on day 1; there was no effect on day 4. Less than a 2-fold increase in the midazolam AUC is not considered clinically important.

ARIPiprazole: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.

Asenapine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.

Aspirin, ASA; Butalbital; Caffeine: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep.

However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Aspirin, ASA; Caffeine: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Aspirin, ASA; Caffeine; Orphenadrine: (Moderate) Additive CNS depressant effects may be seen with combination use of orphenadrine and anxiolytics, sedatives, and hypnotics. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Aspirin, ASA; Carisoprodol; Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Carisoprodol can cause additive CNS depression if used concomitantly with other CNS depressants.
Aspirin, ASA; oxyCODONE: (Major) Concomitant use of oxycodone with zolpidem may lead to additive respiratory and/or CNS depression. Hypotension, profound sedation, coma, respiratory depression, or death may occur. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit

drugs. If zolpidem is used concurrently with oxycodone, a reduced dosage of oxycodone and/or zolpidem is recommended; use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor for sedation and respiratory depression.
Atazanavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Atazanavir; Cobicistat: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%. (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Atropine; Difenoxin: (Moderate) Concurrent administration of diphenoxylate/difenoxin with anxiolytics, sedatives, and hypnotics can potentiate the CNS-depressant effects of diphenoxylate/difenoxin. Use caution during coadministration.
atypical antipsychotic: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Azelastine: (Moderate) An enhanced CNS depressant effect may occur when azelastine is combined with CNS depressants including anxiolytics, sedatives, and hypnotics.
Azelastine; Fluticasone: (Moderate) An enhanced CNS depressant effect may occur when azelastine is combined with CNS depressants including anxiolytics, sedatives, and hypnotics.
Baclofen: (Moderate) Concurrent use of baclofen and CNS depressants such as certain sedatives or hypnotics can cause additive CNS depression. A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during coadministration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Barbiturates: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo

brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Belladonna; Opium: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Benzhydrocodone; Acetaminophen: (Major) Concomitant use of opioid agonists with zolpidem may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If benzhydrocodone is initiated in a patient taking zolpidem, reduce initial dosage and titrate to clinical response. If zolpidem is initiated a patient taking an opioid agonist, use a lower initial dose of zolpidem and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Benzodiazepines: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Benztropine: (Moderate) CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase the sedative effects of benztropine.
Bexarotene: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as bexarotene. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.

Brexpiprazole: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Brimonidine: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of anxiolytics, sedatives, and hypnotics.
Brimonidine; Brinzolamide: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of anxiolytics, sedatives, and hypnotics.
Brimonidine; Timolol: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of anxiolytics, sedatives, and hypnotics.
Brompheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Brompheniramine; Dextromethorphan; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Brompheniramine; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Brompheniramine; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Brompheniramine; Pseudoephedrine; Dextromethorphan: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Buprenorphine: (Major) Reserve concomitant prescribing of buprenorphine and zolpidem for use in patients in whom alternate treatment options are inadequate. Limit dosages and durations to the minimum required and monitor patients closely for respiratory depression and sedation. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Gradually tapering a patient off other CNS depressants or decreasing to

the lowest effective dose is preferred in most cases of patients being treated for opioid use disorder. If concomitant use is necessary, consider prescribing naloxone for the emergency treatment of opioid overdose. Concomitant use can increase the risk of hypotension, respiratory depression, profound sedation, coma, and death.
Buprenorphine; Naloxone: (Major) Reserve concomitant prescribing of buprenorphine and zolpidem for use in patients in whom alternate treatment options are inadequate. Limit dosages and durations to the minimum required and monitor patients closely for respiratory depression and sedation. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Gradually tapering a patient off other CNS depressants or decreasing to the lowest effective dose is preferred in most cases of patients being treated for opioid use disorder. If concomitant use is necessary, consider prescribing naloxone for the emergency treatment of opioid overdose. Concomitant use can increase the risk of hypotension, respiratory depression, profound sedation, coma, and death.
Butalbital; Acetaminophen: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Butalbital; Acetaminophen; Caffeine: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-

containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Butalbital; Acetaminophen; Caffeine; Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Butalbital; Aspirin; Caffeine; Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit

the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer. (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Butorphanol: (Moderate) Concomitant use of butorphanol with other CNS depressants can potentiate the effects of butorphanol on respiratory depression, CNS depression (e.g., dizziness, impaired mental function), and sedation. Butorphanol should be used cautiously in any patient receiving these agents, which may include zolpidem. A dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended in patients receiving concomitant CNS depressants. A reduction in dose of the CNS depressant may also be needed. Concurrent use Intermezzo with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended.
Caffeine: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy

subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Caffeine; Sodium Benzoate: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Calcium, Magnesium, Potassium, Sodium Oxybates: (Contraindicated) Sodium oxybate should not be used in combination with CNS depressant anxiolytics, sedatives, and hypnotics or other sedative CNS depressant drugs. Specifically, sodium oxybate use is contraindicated in patients being treated with sedative hypnotic drugs. Sodium oxybate (GHB) has the potential to impair cognitive and motor skills. For example, the concomitant use of barbiturates and benzodiazepines increases sleep duration and may contribute to rapid onset, pronounced CNS depression, respiratory depression, or coma when combined with sodium oxybate.
Cannabidiol: (Moderate) Monitor for excessive sedation and somnolence during coadministration. CNS depressants can potentiate the effects of cannabidiol.
Capsaicin; Metaxalone: (Moderate) Concomitant administration of metaxalone with other CNS depressants, such as certain sedatives and hypnotics, can potentiate the sedative effects of either agent. A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during co-administration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
carBAMazepine: (Major) Concurrent use of zolpidem with potent CYP3A4 inducers, such as carbamazepine, should be avoided if possible because decreased plasma concentrations of zolpidem are possible and efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamics effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Carbidopa; Levodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.

Carbidopa; Levodopa; Entacapone: (Major) COMT inhibitors such as entacapone and tolcapone should be given cautiously with other agents that cause CNS depression, including zolpidem, due to the possibility of additive sedation. If concurrent use is necessary, monitor for additive side effects. A reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Carbinoxamine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Cariprazine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Carisoprodol: (Moderate) Carisoprodol can cause additive CNS depression if used concomitantly with other CNS depressants.
Celecoxib; Tramadol: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. A dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended in patients receiving concomitant CNS depressants.
Cenobamate: (Moderate) Monitor for excessive sedation and somnolence during coadministration of cenobamate and zolpidem. Concurrent use increases the risk for CNS depression and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Patients should be instructed to contact their provider immediately if these symptoms or behaviors occur. Additionally, monitor for reduced therapeutic response to zolpidem as zolpidem exposure may be decreased. Zolpidem is a CYP3A4 substrate; cenobamate is a moderate CYP3A4 inducer.
Ceritinib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with ceritinib is necessary. Zolpidem is a CYP3A4 substrate and ceritinib is a strong CYP3A4

inhibitor. Coadministration with another strong CYP3A4 inhibitor increased the AUC of zolpidem by 34% to 70%.
Cetirizine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of cetirizine and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Cetirizine; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of cetirizine and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlophedianol; Dexbrompheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlophedianol; Dexchlorpheniramine; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chloramphenicol: (Moderate) Consider decreasing the dose of zolpidem if coadministration with chloramphenicol is necessary. Zolpidem is a CYP3A4 substrate and chloramphenicol is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Chlorcyclizine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
chlordiazePOXIDE: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
chlordiazePOXIDE; Amitriptyline: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the

dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
chlordiazePOXIDE; Clidinium: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Chlorpheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Dextromethorphan: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk

for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Dextromethorphan; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; HYDROcodone: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Ibuprofen; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Chlorpheniramine; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
chlorproMAZINE: (Moderate) Phenothiazines are CNS depressant drugs that may have

cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Chlorzoxazone: (Moderate) Concurrent use of chlorzoxazone and CNS depressants such as certain sedatives or hypnotics can cause additive CNS depression. A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during co-administration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Clarithromycin: (Moderate) Consider decreasing the dose of zolpidem if coadministration with clarithromycin is necessary. Zolpidem is a CYP3A4 substrate and clarithromycin is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Clemastine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
cloBAZam: (Moderate) Concomitant administration of clobazam with other CNS depressant drugs including sedatives and hypnotics, can potentiate the CNS effects (i.e., increased sedation or respiratory depression) of either agent.
clomiPRAMINE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
clonazePAM: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Clorazepate: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either

agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
cloZAPine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Cobicistat: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Codeine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Codeine; Dexbrompheniramine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Codeine; guaiFENesin: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g.,

driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Codeine; guaiFENesin; Pseudoephedrine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Codeine; Phenylephrine; Promethazine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Codeine; Promethazine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid

pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
(Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
COMT inhibitors: (Major) COMT inhibitors such as entacapone and tolcapone should be given cautiously with other agents that cause CNS depression, including zolpidem, due to the possibility of additive sedation. If concurrent use is necessary, monitor for additive side effects. A reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Cyclobenzaprine: (Moderate) Cyclobenzaprine may cause additive CNS depression if used concomitantly with other CNS depressants, such as anxiolytics, sedatives, and hypnotics. Combination therapy can cause additive effects of sedation and dizziness, which can impair the patient's ability to undertake tasks requiring mental alertness. Dosage adjustments of either or both medications may be necessary.
Cyproheptadine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Dabrafenib: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as dabrafenib. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Dantrolene: (Moderate) Simultaneous use of dantrolene and other CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase CNS depression (e.g., drowsiness). A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during co-administration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Daridorexant: (Major) Use of daridorexant with other sedatives and hypnotics should

generally be avoided due to duplication of treatments and due to the additive CNS depressant and complex sleep-related behaviors that may occur. While anxiolytic medications may be used concurrently with daridorexant, a reduction in dose of one or both agents may be needed.
Darunavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Darunavir; Cobicistat: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%. (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Darunavir; Cobicistat; Emtricitabine; Tenofovir alafenamide: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%. (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Deferasirox: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as deferasirox. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Desipramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Desvenlafaxine: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and antidepressants. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. The interaction is thought to be pharmacodynamic in nature; therefore, a similar reaction is possible with

dexvenlafaxine. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction.
Deutetrabenazine: (Moderate) Advise patients that concurrent use of deutetrabenazine and drugs that can cause CNS depression, such as zolpidem, may have additive effects and worsen drowsiness or sedation.
Dexbrompheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Dexbrompheniramine; Dextromethorphan; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Dexbrompheniramine; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Dexchlorpheniramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Dexchlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
dexmedeTOMIDine: (Moderate) Co-administration of dexmedetomidine with anxiolytics, sedatives, and hypnotics is likely to lead to an enhancement of CNS depression.
Dextromethorphan; diphenhydrAMINE; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
diazePAM: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For

Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Dicyclomine: (Moderate) Dicyclomine can cause drowsiness, so it should be used cautiously in patients receiving CNS depressants like anxiolytics, sedatives, and hypnotics.
Difelikefalin: (Moderate) Monitor for dizziness, somnolence, mental status changes, and gait disturbances if concomitant use of difelikefalin with CNS depressants is necessary. Concomitant use may increase the risk for these adverse reactions.
dimenhyDRINATE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
diphenhydrAMINE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
diphenhydrAMINE; Ibuprofen: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
diphenhydrAMINE; Naproxen: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
diphenhydrAMINE; Phenylephrine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Diphenoxylate; Atropine: (Moderate) Concurrent administration of diphenoxylate/difenoxin with anxiolytics, sedatives, and hypnotics can potentiate the CNS-depressant effects of diphenoxylate/difenoxin. Use caution during coadministration.
Doxepin: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease

in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Doxylamine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Doxylamine; Pyridoxine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
droNABinol: (Moderate) Concomitant use of dronabinol with other CNS depressants can potentiate the effects of dronabinol on respiratory depression. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Dronedarone: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of dronedarone, a moderate CYP3A4 inhibitor, since CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
droPERidol: (Moderate) Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and ethanol or other CNS depressants like droperidol than with zolpidem alone. Other CNS depressant drugs may also have cumulative sedative effects when administered concurrently and they should be used cautiously with zolpidem. A dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended in patients receiving concomitant CNS depressants. A reduction in dose of droperidol may also be needed.
Duvelisib: (Moderate) Monitor for an increase in zolpidem-related adverse reactions, including excess sedation, if coadministration with duvelisib is necessary. A dose reduction of zolpidem may be necessary. Zolpidem is a CYP3A4 substrate and duvelisib is a moderate CYP3A4 inhibitor. There is evidence of an increase in pharmacodynamic effects and systemic exposure of zolpidem during coadministration with some potent inhibitors of CYP3A4.
Efavirenz: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as efavirenz. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Efavirenz; Emtricitabine; Tenofovir Disoproxil Fumarate: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate

CYP3A4 inducers, such as efavirenz. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Efavirenz; lamiVUDine; Tenofovir Disoproxil Fumarate: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as efavirenz. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Elagolix: (Moderate) Monitor for reduced therapeutic response to zolpidem is coadministration with elagolix is necessary. Zolpidem exposure may be decreased. Zolpidem is a CYP3A4 substrate; elagolix is a weak to moderate CYP3A4 inducer.
Elagolix; Estradiol; Norethindrone acetate: (Moderate) Monitor for reduced therapeutic response to zolpidem is coadministration with elagolix is necessary. Zolpidem exposure may be decreased. Zolpidem is a CYP3A4 substrate; elagolix is a weak to moderate CYP3A4 inducer.
Elbasvir; Grazoprevir: (Moderate) The primary isoenzyme responsible for the metabolism of zolpidem is CYP3A4. The effect of weak CYP3A4 inhibitors, such as elbasvir; grazoprevir, on zolpidem exposure is not known. Until further information is available, it is advisable to monitor for zolpidem-related CNS effects when this combination is administered. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
Elvitegravir; Cobicistat; Emtricitabine; Tenofovir Alafenamide: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Elvitegravir; Cobicistat; Emtricitabine; Tenofovir Disoproxil Fumarate: (Moderate) Consider decreasing the dose of zolpidem if coadministration with cobicistat is necessary. Zolpidem is a CYP3A4 substrate and cobicistat is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Encorafenib: (Major) Coadministration of zolpidem with encorafenib is not recommended as concurrent use may decrease plasma concentrations of zolpidem. Zolpidem is a CYP3A substrate and encorafenib is a strong CYP3A inducer. Coadministration with another strong CYP3A inducer decreased zolpidem exposure by 73% and significantly reduced the pharmacodynamic effects of zolpidem.
Entacapone: (Major) COMT inhibitors such as entacapone and tolcapone should be given

cautiously with other agents that cause CNS depression, including zolpidem, due to the possibility of additive sedation. If concurrent use is necessary, monitor for additive side effects. A reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Enzalutamide: (Major) Coadministration of zolpidem with enzalutamide is not recommended due to decreased plasma concentrations of zolpidem. Zolpidem is a CYP3A4 substrate and enzalutamide is a strong CYP3A4 inducer. Coadministration with another strong CYP3A4 inducer decreased zolpidem exposure by 73% and significantly reduced the pharmacodynamic effects of zolpidem.
Ergotamine; Caffeine: (Minor) Caffeine is a central nervous system (CNS) stimulant. Patients taking medications for sleep, such as zolpidem should avoid caffeine-containing medications, dietary supplements, foods, and beverages within the hours close to bedtime. Patients should be encouraged to avoid excessive total daily caffeine intake, as part of proper sleep hygiene, since caffeine intake can interfere with proper sleep. However, in healthy subjects (without insomnia) in a pharmacokinetic study, coadministration of caffeine at a dosage of 150 to 300 mg with zolpidem did not counteract the sedative effects of a single 10 mg dose of zolpidem.
Esketamine: (Major) Use of zolpidem during treatment with esketamine may increase sedation and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Instruct patients to contact their provider immediately if these symptoms or behaviors occur and not to drive or engage in other activities requiring alertness until the next day after a restful sleep.
Eslicarbazepine: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as eslicarbazepine. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamics effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Estazolam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Ethanol: (Major) Advise patients not to use zolpidem if they drank alcohol that evening or before bed. An additive adverse effect on psychomotor performance between alcohol and zolpidem has been demonstrated. The risk of next-day psychomotor impairment, including impaired driving, is increased if zolpidem is taken with alcohol.
Etomidate: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may

increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Etravirine: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as etravirine. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamics effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Fedratinib: (Moderate) Monitor for an increase in zolpidem-related adverse reactions, including excess sedation, if coadministration with fedratinib is necessary. A dose reduction of zolpidem may be necessary. Zolpidem is a CYP3A4 substrate and fedratinib is a moderate CYP3A4 inhibitor. There is evidence of an increase in pharmacodynamic effects and systemic exposure of zolpidem during coadministration with some potent inhibitors of CYP3A4.
Fenfluramine: (Moderate) Monitor for excessive sedation and somnolence during coadministration of fenfluramine and zolpidem. Concurrent use increases the risk for CNS depression and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake).
fentaNYL: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Flumazenil: (Major) Flumazenil, a benzodiazepine antagonist, can reduce the sedative hypnotic effects of zolpidem. Flumazenil and zolpidem are pharmacological opposites and should not be used together therapeutically.
fluPHENAZine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Flurazepam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either

agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
fluvoxaMINE: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of fluvoxamine, since the metabolism of zolpidem may be decreased by CYP3A4, CYP1A2, and CYP2C9 inhibition of fluvoxamine. There is evidence of an increase in pharmacodynamic effects and systemic exposure of zolpidem when the drug is co-administered with some potent inhibitors of CYP3A4, such as azole antifungals. In addition, disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and SSRIs including fluvoxamine. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction. In one case report, an elderly female patient experienced visual hallucinations and amnesia of the events three days after an increase in fluvoxamine dose to 50 mg three times daily after stabilization on zolpidem 10 mg/night, and fluvoxamine 50 mg twice daily. The mechanism for the interaction is unknown but may be pharmacodynamic in nature and/or related to increased zolpidem exposure from fluvoxamine inhibition of CYP3A4.
Food: (Major) Advise patients to avoid cannabis use while taking CNS depressants due to the risk for additive CNS depression and potential for other cognitive adverse reactions.
Fosamprenavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Foscarbidopa; Foslevodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Fosphenytoin: (Major) Avoid concurrent use of zolpidem with potent CYP3A4 inducers, such as hydantoins, if possible due to the potential for decreased zolpidem exposure and potential reduction of efficacy. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration with

another potent CYP3A4 inducer. Consider if an alternative sleep agent would be appropriate for the patient.
Gabapentin: (Major) Initiate gabapentin at the lowest recommended dose and monitor patients for symptoms of sedation and somnolence during coadministration of gabapentin and zolpidem. The recommended dose of sublingual zolpidem tablets is 1.75 mg/night in patients receiving concomitant CNS depressants. Concomitant use of gabapentin with zolpidem may cause additive CNS depression and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Educate patients about the risks and symptoms of excessive CNS depression. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur.
General anesthetics: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Haloperidol: (Moderate) Haloperidol can potentiate the actions of other CNS depressants such as anxiolytics, sedatives, and hypnotics, and they should be used cautiously in combination.
Homatropine; HYDROcodone: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Hydantoins: (Major) Avoid concurrent use of zolpidem with potent CYP3A4 inducers, such as hydantoins, if possible due to the potential for decreased zolpidem exposure and potential reduction of efficacy. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration with another potent CYP3A4 inducer. Consider if an alternative sleep agent would be appropriate for the patient.
HYDROcodone: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain

medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
HYDROcodone; Ibuprofen: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Avoid prescribing opioid cough medications in patients taking zolpidem. Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
HYDROmorphone: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
hydrOXYzine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Idelalisib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with idelalisib is necessary. Zolpidem is a CYP3A4 substrate and idelalisib is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Iloperidone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Imatinib: (Major) It is advisable to closely monitor zolpidem tolerability and safety during co-administration of CYP3A4 inhibitors, such as imatinib, STI-571, and consider using a lower dose of zolpidem to minimize the potential for adverse CNS effects. CYP3A4 is the

primary isoenzyme responsible for zolpidem metabolism, and there is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
Imipramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Isavuconazonium: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of isavuconazonium, a moderate CYP3A4 inhibitor, since CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
Isocarboxazid: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of anxiolytics, sedatives, and hypnotics and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
Isoflurane: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Itraconazole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with itraconazole is necessary. Zolpidem is a CYP3A4 substrate and itraconazole is a strong CYP3A4 inhibitor. Coadministration with itraconazole increased the AUC of zolpidem by 34%.
Ketamine: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Ketoconazole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with ketoconazole is necessary. Zolpidem is a CYP3A4 substrate and ketoconazole is a strong CYP3A4 inhibitor. Coadministration with ketoconazole increased the AUC (exposure) of zolpidem by 70% and also increased the pharmacodynamic effects of zolpidem (e.g., sedation and psychomotor impairment).
Lansoprazole; Amoxicillin; Clarithromycin: (Moderate) Consider decreasing the dose of zolpidem if coadministration with clarithromycin is necessary. Zolpidem is a CYP3A4

substrate and clarithromycin is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Lasmiditan: (Moderate) Monitor for excessive sedation and somnolence during coadministration of lasmiditan and zolpidem. Concurrent use increases the risk for CNS depression and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Patients should be instructed to contact their provider immediately if these symptoms or behaviors occur.
Lefamulin: (Moderate) Monitor for an increase in zolpidem-related adverse reactions, including excess sedation, if coadministration with oral lefamulin is necessary. A dose reduction of zolpidem may be necessary. Zolpidem is a CYP3A4 substrate and oral lefamulin is a moderate CYP3A4 inhibitor; an interaction is not expected with intravenous lefamulin. There is evidence of an increase in pharmacodynamic effects and systemic exposure of zolpidem during coadministration with some potent inhibitors of CYP3A4.
Lemborexant: (Moderate) Use of lemborexant with other sedatives and hypnotics should generally be avoided due to duplication of treatments and due to the additive CNS depressant and complex sleep-related behaviors that may occur. While anxiolytic medications may be used concurrently with lemborexant, a reduction in dose of one or both agents may be needed.
Letermovir: (Moderate) Plasma concentrations and pharmacodynamic effect of zolpidem could be increased when administered concurrently with letermovir. Closely monitor for adverse events and consider zolpidem dose reductions if appropriate in patients also receiving cyclosporine because the magnitude of the interaction may increase. Zolpidem is primarily metabolized by CYP3A4. Letermovir is a moderate inhibitor of CYP3A4; however, when given with cyclosporine, the combined effect on CYP3A4 substrates may be similar to a strong CYP3A4 inhibitor. Concurrent use with a strong CYP3A4 inhibitor increased the zolpidem Cmax and AUC by 30% and 70%, respectively. In addition, concurrent use prolonged zolpidem half-life by 30% and increased the pharmacodynamic effects of zolpidem.
Levocetirizine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of cetirizine and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Levodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Levoketoconazole: (Moderate) Consider decreasing the dose of zolpidem if

coadministration with ketoconazole is necessary. Zolpidem is a CYP3A4 substrate and ketoconazole is a strong CYP3A4 inhibitor. Coadministration with ketoconazole increased the AUC (exposure) of zolpidem by 70% and also increased the pharmacodynamic effects of zolpidem (e.g., sedation and psychomotor impairment).
Levomilnacipran: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and some SNRI antidepressants. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. The interaction is thought to be pharmacodynamic in nature; therefore, a similar reaction is possible with levomilnacipran. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction.
Levorphanol: (Major) Concomitant use of levorphanol with other CNS depressants such as zolpidem can potentiate the effects of levorphanol on respiration, blood pressure, and alertness. Severe hypotension, respiratory depression, profound sedation, or coma may occur. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. When concomitant treatment is necessary, reduce the dose of 1 or both drugs. The initial dose of levorphanol should be reduced by approximately 50% or more when levorphanol is used with another drug that may depress respiration. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Lithium: (Moderate) Because lithium has the potential to impair cognitive and motor skills, caution is advisable during concurrent use of other medications with centrally-acting effects including anxiolytics, sedatives, and hypnotics.
Lofexidine: (Moderate) Monitor for additive sedation during coadministration of lofexidine and anxiolytics, sedatives, and hypnotics. Lofexidine can potentiate the effects of CNS depressants. Patients should be advised to avoid driving or performing any other tasks requiring mental alertness until the effects of the combination are known.
Lonafarnib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with lonafarnib is necessary. Concurrent use may increase zolpidem exposure. Zolpidem is a CYP3A4 substrate and lonafarnib is a strong CYP3A4 inhibitor. Coadministration with other strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Lopinavir; Ritonavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
LORazepam: (Major) Concomitant administration of benzodiazepines with zolpidem can

potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Loxapine: (Moderate) CNS depressant drugs, including loxapine, may have cumulative effects when administered concurrently and they should be used cautiously with zolpidem. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Lumacaftor; Ivacaftor: (Major) Concurrent use of zolpidem with potent CYP3A4 inducers, such as lumacaftor; ivacaftor, should be avoided if possible because decreased plasma concentrations of zolpidem are possible and efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Lumacaftor; Ivacaftor: (Major) Concurrent use of zolpidem with potent CYP3A4 inducers, such as lumacaftor; ivacaftor, should be avoided if possible because decreased plasma concentrations of zolpidem are possible and efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Lumateperone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Lurasidone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Maprotiline: (Moderate) CNS depressants should be combined cautiously with maprotiline because they could cause additive depressant effects and possible respiratory depression or hypotension.
Meclizine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Melatonin: (Major) Pharmacodynamic interactions often occur when sedative agents are

used together. Avoid combining melatonin with zolpidem. In a clinical trial, there was clear evidence for a transitory pharmacodynamic interaction between melatonin and zolpidem one hour following co-dosing. Concomitant administration resulted in increased impairment of attention, memory and coordination compared to zolpidem alone. Use of more than one agent for hypnotic purposes may increase the risk for over-sedation, CNS effects, or sleep-related behaviors. Be alert for unusual changes in moods or behaviors. Patients reporting unusual sleep-related behaviors likely should discontinue melatonin use.
Meperidine: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Metaxalone: (Moderate) Concomitant administration of metaxalone with other CNS depressants, such as certain sedatives and hypnotics, can potentiate the sedative effects of either agent. A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during co-administration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Methadone: (Major) Concomitant use of methadone with zolpidem can lead to additive respiratory depression, hypotension, profound sedation, or coma. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. Methadone should be used with caution and in reduced dosages if used concurrently with a CNS depressant; in opioid-naive adults, use an initial methadone dose of 2.5 mg every 12 hours. Also consider a using a lower dose of zolpidem. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor patients for sedation and respiratory depression.
Methocarbamol: (Moderate) Methocarbamol may cause additive CNS depression if used concomitantly with other CNS depressants. Dosage reduction of one or both agents may be necessary.
Methohexital: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving,

are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Methscopolamine: (Moderate) CNS depression can be increased when methscopolamine is combined with other CNS depressants such as any anxiolytics, sedatives, and hypnotics.
Metoclopramide: (Minor) Combined use of metoclopramide and other CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase possible sedation.
metyraPONE: (Moderate) Metyrapone may cause dizziness and/or drowsiness. Other drugs that may also cause drowsiness, such as zolpidem, should be used with caution. Additive drowsiness and/or dizziness is possible.
Midazolam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
miFEPRIStone: (Moderate) Consider decreasing the dose of zolpidem if coadministration with mifepristone is necessary. Concurrent use may increase zolpidem exposure. The clinical significance of this interaction with the short-term use of mifepristone for termination of pregnancy is unknown. Zolpidem is a CYP3A substrate and mifepristone is a strong CYP3A inhibitor. Coadministration with other strong CYP3A inhibitors increased the AUC of zolpidem by 34% to 70%.
Milnacipran: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and some SNRI-type antidepressants. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. The interaction is thought to be pharmacodynamic in nature; therefore, a similar reaction is possible with levomilnacipran or milnacipran. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction.

Mirtazapine: (Moderate) Consistent with the pharmacology of mirtazapine and the drug's side effect profile, additive effects may occur with other CNS-active agents, including anxiolytics, sedatives, and hypnotics.
Mitotane: (Major) Concurrent use of zolpidem with potent CYP3A4 inducers, such as mitotane, should be avoided if possible because decreased plasma concentrations of zolpidem are possible and efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamics effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer. Additionally, mitotane can cause sedation, lethargy, vertigo, and other CNS adverse reactions; additive CNS effects may occur when mitotane is given with zolpidem.
Modafinil: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as modafinil. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamics effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Molindone: (Moderate) Additive CNS-depressant effects may occur with zolpidem and molindone. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Monoamine oxidase inhibitors: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of anxiolytics, sedatives, and hypnotics and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
Morphine: (Moderate) Concomitant use of morphine with zolpidem can potentiate the effects of morphine on respiration, blood pressure, and alertness. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. If zolpidem is used concurrently with morphine, a reduced dosage of morphine and/or zolpidem is recommended; for extended-release products, start with the lowest possible dose of morphine (i.e., 15 mg PO every 12 hours, extended-release tablets; 30 mg or less PO every 24 hours; extended-release capsules). For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor patients for sedation and respiratory depression.
Nalbuphine: (Moderate) Concomitant use of nalbuphine with other CNS depressants, such as zolpidem, can potentiate the effects of nalbuphine on respiratory depression, CNS depression, and sedation. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics,

including other zolpidem products, at bedtime or the middle of the night is not recommended.
Nefazodone: (Moderate) Consider decreasing the dose of zolpidem if coadministration with nefazodone is necessary. Zolpidem is a CYP3A4 substrate and nefazodone is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Nelfinavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Netupitant, Fosnetupitant; Palonosetron: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of netupitant, a moderate CYP3A4 inhibitor, since CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
Nilotinib: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of nilotinib, a moderate CYP3A4 inhibitor, since CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem when the drug is co-administered with some potent inhibitors of CYP3A4, such as azole antifungals. Concurrent administration of nilotinib and midazolam, a CYP3A4 substrate, increased midazolam exposure by 30%.
Nirmatrelvir; Ritonavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Nortriptyline: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
OLANZapine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
OLANZapine; FLUoxetine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for

additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
OLANZapine; Samidorphan: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Oliceridine: (Major) Concomitant use of oliceridine with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of oliceridine with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Omeprazole; Amoxicillin; Rifabutin: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as rifabutin. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
Opicapone: (Major) COMT inhibitors such as entacapone and tolcapone should be given cautiously with other agents that cause CNS depression, including zolpidem, due to the possibility of additive sedation. If concurrent use is necessary, monitor for additive side effects. A reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Oritavancin: (Moderate) Zolpidem is metabolized by CYP3A4; oritavancin is a weak CYP3A4 inducer. Plasma concentrations and efficacy of zolpidem may be reduced if these drugs are administered concurrently.
Orphenadrine: (Moderate) Additive CNS depressant effects may be seen with combination use of orphenadrine and anxiolytics, sedatives, and hypnotics.
Oxazepam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
oxyCODONE: (Major) Concomitant use of oxycodone with zolpidem may lead to additive respiratory and/or CNS depression. Hypotension, profound sedation, coma, respiratory depression, or death may occur. In addition, sleep-related behaviors, such as sleep-

driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. If zolpidem is used concurrently with oxycodone, a reduced dosage of oxycodone and/or zolpidem is recommended; use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor for sedation and respiratory depression.
oxyMORphone: (Major) Concomitant use of oxymorphone with zolpidem may produce additive CNS depressant effects. Hypotension, profound sedation, coma, respiratory depression, or death may occur. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Prior to concurrent use, assess the level of tolerance to CNS depression that has developed, the duration of use, and the patient's overall response to treatment. Consider the patient's use of alcohol or illicit drugs. If zolpidem is used concurrently with oxymorphone, a reduced dosage of oxymorphone (1/3 to 1/2 of the usual dose) and/or zolpidem is recommended. If the extended-release oxymorphone tablets are used concurrently with a CNS depressant, it is recommended to use an initial oxymorphone ER dosage of 5 mg PO every 12 hours. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Monitor for sedation or respiratory depression.
Palbociclib: (Moderate) Monitor for an increase in sedation and other zolpidem-related adverse reactions if coadministration with palbociclib is necessary. Zolpidem is a CYP3A4 substrate and palbociclib is a weak, time-dependent CYP3A4 inhibitor.
Paliperidone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Papaverine: (Moderate) Papaverine is a benzylisoquinoline alkaloid of opium and may have synergistic effects with potent CNS depressants such as anxiolytics, sedatives, and hypnotics, which could lead to enhanced sedation.
PARoxetine: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and antidepressants. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. The interaction is thought to be pharmacodynamic in nature; therefore, a similar reaction is possible with paroxetine. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction.
PAZOPanib: (Moderate) Pazopanib is a weak inhibitor of CYP3A4. Coadministration of

pazopanib and zolpidem, a CYP3A4 substrate, may cause an increase in systemic concentrations of zolpidem. Use caution when administering these drugs concomitantly.
Pentazocine; Naloxone: (Moderate) Concomitant use of pentazocine with zolpidem can potentiate respiratory depression, CNS depression, and sedation. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone. Pentazocine should be used cautiously in any patient receiving zolpidem. If concurrent use is necessary, a dose reduction of one or both medications may be required. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
PENTobarbital: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Perampanel: (Moderate) Co-administration of perampanel with CNS depressants, including ethanol, may increase CNS depression. The combination of perampanel (particularly at high doses) with ethanol has led to decreased mental alertness and ability to perform complex tasks (such as driving), as well as increased levels of anger, confusion, and depression; similar reactions should be expected with concomitant use of other CNS depressants, such as zolpidem.
Perphenazine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Perphenazine; Amitriptyline: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease

in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness. (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Phenelzine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of anxiolytics, sedatives, and hypnotics and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
PHENobarbital: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
PHENobarbital; Hyoscyamine; Atropine; Scopolamine: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer. (Moderate) Scopolamine may cause dizziness and drowsiness. Concurrent use of scopolamine and CNS depressants can adversely increase the risk of CNS depression.
Phenothiazines: (Moderate) Phenothiazines are CNS depressant drugs that may have

cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Phentermine; Topiramate: (Moderate) Although not specifically studied, coadministration of CNS depressant drugs (e.g., anxiolytics, sedatives, and hypnotics) with topiramate may potentiate CNS depression such as dizziness or cognitive adverse reactions, or other centrally mediated effects of these agents. Monitor for increased CNS effects if coadministering.
Phenytoin: (Major) Avoid concurrent use of zolpidem with potent CYP3A4 inducers, such as hydantoins, if possible due to the potential for decreased zolpidem exposure and potential reduction of efficacy. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration with another potent CYP3A4 inducer. Consider if an alternative sleep agent would be appropriate for the patient.
Pimozide: (Moderate) Additive CNS-depressant effects may occur when the antipsychotic pimozide is combined with zolpidem. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Posaconazole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with posaconazole is necessary. Zolpidem is a CYP3A4 substrate and posaconazole is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Pramipexole: (Moderate) Other CNS depressant drugs, such as pramipexole, may have cumulative effects when administered concurrently with zolpidem and they should be used cautiously with zolpidem. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended.
Pregabalin: (Major) Initiate pregabalin at the lowest recommended dose and monitor patients for symptoms of sedation and somnolence during coadministration of pregabalin and zolpidem. The recommended dose of sublingual zolpidem tablets is 1.75 mg/night in patients receiving concomitant CNS depressants. Concomitant use of pregabalin with zolpidem may cause additive CNS depression and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Educate patients about the risks and symptoms of excessive CNS depression.

Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur.
Primidone: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Prochlorperazine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Promethazine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Promethazine; Dextromethorphan: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Promethazine; Phenylephrine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-

driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Propofol: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Protease inhibitors: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Protriptyline: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Pseudoephedrine; Triprolidine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Quazepam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
QUEtiapine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
quiNINE: (Moderate) It is advisable to closely monitor zolpidem tolerability and safety during concurrent use of quinine, a moderate CYP3A4 inhibitor, since CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism. There is evidence of an increase in pharmacodynamics effects and systemic exposure of zolpidem during co-administration with some potent inhibitors of CYP3A4, such as azole antifungals.
Ramelteon: (Moderate) Ramelteon is a sleep-promoting agent; therefore, additive

pharmacodynamic effects are possible when combining ramelteon with benzodiazepines or other miscellaneous anxiolytics, sedatives, and hypnotics. Pharmacokinetic interactions have been observed with the use of zolpidem. Use of ramelteon 8 mg/day for 11 days and a single dose of zolpidem 10 mg resulted in an increase in the median Tmax of zolpidem of about 20 minutes; exposure to zolpidem was unchanged. Ramelteon use with hypnotics of any kind is considered duplicative therapy and these drugs are generally not co-administered.
Ranolazine: (Moderate) In vitro studies indicate that ranolazine and its O-demethylated metabolite are inhibitors of CYP3A isoenzymes. In theory, ranolazine may inhibit zolpidem CYP3A4 metabolism, potentially leading to increased zolpidem plasma concentrations. Although not studied, excessive sedation and possible respiratory depression may occur.
Rasagiline: (Moderate) The CNS-depressant effects of MAOIs can be potentiated with concomitant administration of other drugs known to cause CNS depression including buprenorphine, butorphanol, dronabinol, THC, nabilone, nalbuphine, and anxiolytics, sedatives, and hypnotics. Use these drugs cautiously with MAOIs; warn patients to not drive or perform other hazardous activities until they know how a particular drug combination affects them. In some cases, the dosages of the CNS depressants may need to be reduced.
Remifentanil: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Remimazolam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Ribociclib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with ribociclib is necessary. Concurrent use may increase zolpidem exposure. Zolpidem is a CYP3A substrate and ribociclib is a strong CYP3A inhibitor. Coadministration with other strong CYP3A inhibitors increased the AUC of zolpidem by 34% to 70%.

Ribociclib; Letrozole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with ribociclib is necessary. Concurrent use may increase zolpidem exposure. Zolpidem is a CYP3A substrate and ribociclib is a strong CYP3A inhibitor. Coadministration with other strong CYP3A inhibitors increased the AUC of zolpidem by 34% to 70%.
Rifabutin: (Moderate) It is advisable to closely monitor for reductions in zolpidem efficacy during co-administration of moderate CYP3A4 inducers, such as rifabutin. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin, a potent CYP3A4 inducer.
rifAMPin: (Major) Concurrent use of zolpidem with potent CYP3A4 inducers, such as rifampin, is not recommended. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of a significant decrease in systemic exposure and pharmacodynamic effects of zolpidem during co-administration of rifampin. During one small drug interaction study of healthy females, a single 10 mg dose of zolpidem co-administered with rifampin at steady state levels resulted in a 73%, 58%, and 36% decrease in the AUC, Cmax, and half-life, respectively, of zolpidem; this decrease in exposure resulted in significant reductions in the pharmacodynamic effects of zolpidem.
Rifapentine: (Major) Coadministration of zolpidem with rifapentine is not recommended as concurrent use may decrease plasma concentrations of zolpidem. Zolpidem is a CYP3A4 substrate and rifapentine is a strong CYP3A4 inducer. Coadministration with another strong CYP3A4 inducer decreased zolpidem exposure by 73% and significantly reduced the pharmacodynamic effects of zolpidem.
risperiDONE: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Ritonavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
rOPINIRole: (Moderate) Concomitant use of ropinirole with other CNS depressants, such as zolpidem, can potentiate the sedation effects of ropinirole. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Rotigotine: (Moderate) A reduction in the dose of anxiolytics, sedatives, hypnotics and concomitantly administered dopamine agonists with sedative properties (e.g., ropinirole, pramipexole, rotigotine, apomorphine) should be considered to minimize additive sedative effects. In addition, the risk of next-day psychomotor impairment is increased

during co-administration, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Rufinamide: (Minor) Rufinamide is a weak inducer of CYP3A4. In theory, decreased exposure of drugs that are extensively metabolized by CYP3A4, such as zolpidem, may occur during concurrent use with rufinamide.
Safinamide: (Moderate) Dopaminergic medications, including safinamide, may cause a sudden onset of somnolence which sometimes has resulted in motor vehicle accidents. Patients may not perceive warning signs, such as excessive drowsiness, or they may report feeling alert immediately prior to the event. Because of possible additive effects, advise patients about the potential for increased somnolence during concurrent use of safinamide with other sedating medications, such as anxiolytics, sedatives, and hypnotics.
Saquinavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Scopolamine: (Moderate) Scopolamine may cause dizziness and drowsiness. Concurrent use of scopolamine and CNS depressants can adversely increase the risk of CNS depression.
Secobarbital: (Major) Concurrent use of zolpidem with barbiturates should be avoided if possible due to additive CNS depression. Sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than zolpidem alone. Concurrent use of zolpidem with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. Dosage reduction may be required for co-use in some patients. For example, a dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended when used with another CNS depressant. Barbiturates are CYP3A4 enzyme inducers and may cause decreased plasma concentrations of zolpidem; in some patients efficacy may be reduced. CYP3A4 is the primary isoenzyme responsible for zolpidem metabolism, and there is evidence of significant decreases in systemic exposure and pharmacodynamic effects of zolpidem during coadministration of a potent CYP3A4 inducer.
Sedating H1-blockers: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Selegiline: (Moderate) Monitor for excessive sedation and somnolence during coadministration of selegiline with anxiolytics, sedatives, and hypnotics. Concurrent use may result in additive CNS depression.
Sertraline: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and SSRIs including sertraline. The duration

of the visual hallucinations ranged from 30 minutes to 7 hours. Data from a clinical study in which SSRI-treated patients were given immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction. The mechanism for the interaction is thought to be pharmacodynamic in nature. In one study with sertraline, inhibition of zolpidem metabolism occurred when sertraline was chronically coadministered; zolpidem Cmax was significantly higher (43%) and Tmax was significantly decreased (53%).
Sevoflurane: (Moderate) The effects of CNS depressant drugs, such as zolpidem, may increase when administered concurrently with general anesthetics. A temporary dose reduction of the CNS depressant should be considered following administration of general anesthetics. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Sodium Oxybate: (Contraindicated) Sodium oxybate should not be used in combination with CNS depressant anxiolytics, sedatives, and hypnotics or other sedative CNS depressant drugs. Specifically, sodium oxybate use is contraindicated in patients being treated with sedative hypnotic drugs. Sodium oxybate (GHB) has the potential to impair cognitive and motor skills. For example, the concomitant use of barbiturates and benzodiazepines increases sleep duration and may contribute to rapid onset, pronounced CNS depression, respiratory depression, or coma when combined with sodium oxybate.
St. John's Wort, Hypericum perforatum: (Major) Avoid use of St. John's wort with zolpidem is possible; use of St. John's wort is not recommended as concurrent use may decrease plasma concentrations of zolpidem and lead to loss of hypnotic efficacy. Zolpidem is a CYP3A4 substrate and St. John's wort is a strong CYP3A4 inducer. Coadministration with St. John's wort decreased the mean AUC of zolpidem by 30%.
Stiripentol: (Moderate) Monitor for excessive sedation and somnolence during coadministration of stiripentol and zolpidem. CNS depressants can potentiate the effects of stiripentol.
SUFentanil: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Suvorexant: (Moderate) Use of suvorexant with other sedatives and hypnotics should generally be avoided due to duplication of treatments and due to the additive CNS

depressant and complex sleep-related behaviors that may occur. While anxiolytic medications may be used concurrently with suvorexant, a reduction in dose of one or both of these agents may be needed.
Tapentadol: (Major) Concomitant use of opioid agonists with other CNS depressants may cause excessive sedation and somnolence. Limit the use of opioid pain medications with anotehr CNS depressant to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Educate patients about the risks and symptoms of excessive CNS depression.
Temazepam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Tetrabenazine: (Moderate) Concurrent use of tetrabenazine and drugs that can cause CNS depression, such as anxiolytics, sedatives, and hypnotics, can increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, dizziness, and orthostatic hypotension.
Thalidomide: (Major) The use of anxiolytics, sedatives, or hypnotics concomitantly with thalidomide may cause an additive sedative effect and should be avoided. Thalidomide frequently causes drowsiness and somnolence. Dose reductions may be required. Patients should be instructed to avoid situations where drowsiness may be a problem and not to take other medications that may cause drowsiness without adequate medical advice. Advise patients as to the possible impairment of mental and/or physical abilities required for the performance of hazardous tasks, such as driving a car or operating other complex or dangerous machinery.
Thioridazine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Thiothixene: (Moderate) Zolpidem and thiothixene may have cumulative effects on CNS depression when administered concurrently and they should be used together with caution. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem

alone.
Tipranavir: (Moderate) Consider decreasing the dose of zolpidem if coadministration with protease inhibitors is necessary. Zolpidem is a CYP3A4 substrate and protease inhibitors are strong CYP3A4 inhibitors. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
tiZANidine: (Moderate) Monitor for excessive sedation and somnolence during coadministration of tizanidine with anxiolytics, sedatives, and hypnotics. Concurrent use may result in additive CNS depression. A reduction in the dose of these medications may be considered to minimize additive sedative effects, if they occur. With hypnotic medications, the risk of next-day psychomotor impairment is increased during co-administration of other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving.
Tolcapone: (Major) COMT inhibitors such as entacapone and tolcapone should be given cautiously with other agents that cause CNS depression, including zolpidem, due to the possibility of additive sedation. If concurrent use is necessary, monitor for additive side effects. A reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night.
Topiramate: (Moderate) Although not specifically studied, coadministration of CNS depressant drugs (e.g., anxiolytics, sedatives, and hypnotics) with topiramate may potentiate CNS depression such as dizziness or cognitive adverse reactions, or other centrally mediated effects of these agents. Monitor for increased CNS effects if coadministering.
traMADol: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS depression. A dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended in patients receiving concomitant CNS depressants.
Tramadol; Acetaminophen: (Major) Concomitant use of opioid agonists with zolpidem may cause excessive sedation, somnolence, and complex sleep-related behaviors (e.g., driving, talking, eating, or performing other activities while not fully awake). Limit the use of opioid pain medications with zolpidem to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. Instruct patients to contact their provider immediately if sleep-related symptoms or behaviors occur. Educate patients about the risks and symptoms of excessive CNS

depression. A dosage reduction of the Intermezzo brand of sublingual zolpidem tablets to 1.75 mg/night is recommended in patients receiving concomitant CNS depressants.
Tranylcypromine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of anxiolytics, sedatives, and hypnotics and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
traZODone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of trazodone and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Triazolam: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Tricyclic antidepressants: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day. Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Trifluoperazine: (Moderate) Phenothiazines are CNS depressant drugs that may have cumulative effects when administered with other CNS depressant drugs and they should be used cautiously with anxiolytic, sedative, and hypnotics. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension. Additionally, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of hypnotics and other CNS depressants than with use of a hypnotic alone.
Trihexyphenidyl: (Moderate) CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase the sedative effects of trihexyphenidyl.
Trimethobenzamide: (Moderate) The concurrent use of trimethobenzamide with other medications that cause CNS depression, like anxiolytics, sedatives and hypnotics, may potentiate the effects of either trimethobenzamide or the anxiolytics, sedatives and hypnotics.
Trimipramine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of tricyclic antidepressants and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be

necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Coadministration of a tricyclic antidepressant and zolpidem resulted in a 20% decrease in the peak concentrations of the tricyclic antidepressant, but there was an additive effect of decreased alertness.
Triprolidine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of sedating H1-blockers and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Tucatinib: (Moderate) Consider decreasing the dose of zolpidem if coadministration with tucatinib is necessary. Concurrent use may increase zolpidem exposure. Zolpidem is a CYP3A4 substrate and tucatinib is a strong CYP3A4 inhibitor. Coadministration with other strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Valerian, Valeriana officinalis: (Major) Patients who are taking sedative/hypnotic drugs should generally avoid concomitant administration of valerian. Any substances that act on the CNS, including sedatives and hypnotics, may theoretically interact with valerian, Valeriana officinalis. The valerian derivative, dihydrovaltrate, binds at barbiturate binding sites; valerenic acid has been shown to inhibit enzyme-induced breakdown of GABA in the brain; the non-volatile monoterpenes (valepotriates) have sedative activity. These interactions are probably pharmacodynamic in nature. There is a possibility of interaction with valerian at normal prescription dosages of sedatives and hypnotics.
Valproic Acid, Divalproex Sodium: (Moderate) A probable interaction between zolpidem and valproic acid resulted in somnambulism (sleep walking) in one case report. A 47 year old patient with a history of bipolar disorder was receiving citalopram (40 mg once daily) and zolpidem (5 mg at bedtime). Manic symptoms developed during treatment and he received valproic acid. Somnambulism developed 2 days after the valproic acid was initiated. The sleep walking stopped after the valproic acid was discontinued and with a rechallenge the symptoms reappeared. It is not known if this interaction is of a pharmacokinetic or pharmacodynamic nature. Somnambulism has also been reported as a rare side effect of zolpidem when used without interacting medications; however this patient did not experience sleep walking with zolpidem monotherapy or with valproic acid monotherapy.
Vemurafenib: (Moderate) Vemurafenib is an inducer of CYP3A4 and decreased plasma concentrations of drugs metabolized by this enzyme, such as zolpidem, could be expected with concurrent use. Use caution, and monitor therapeutic effects of zolpidem when coadministered with vemurafenib.
Venlafaxine: (Moderate) Disorientation, delusions, or hallucinations have been reported rarely during co-administration of zolpidem and antidepressants. The duration of the visual hallucinations has ranged from 30 minutes to 7 hours. The interaction is thought to be pharmacodynamic in nature; therefore, a similar reaction is possible with venlafaxine. Data from a clinical study in which SSRI-treated patients were given

immediate-release zolpidem revealed that four of the seven discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction.
Vigabatrin: (Moderate) Vigabatrin may cause somnolence and fatigue. Drugs that can cause CNS depression, if used concomitantly with vigabatrin, may increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, and dizziness. Caution should be used when vigabatrin is given with anxiolytics, sedatives, and hypnotics.
Vilazodone: (Moderate) Due to the CNS effects of vilazodone, caution should be used when vilazodone is given in combination with other centrally acting medications such as anxiolytics, sedatives, and hypnotics.
Vonoprazan; Amoxicillin; Clarithromycin: (Moderate) Consider decreasing the dose of zolpidem if coadministration with clarithromycin is necessary. Zolpidem is a CYP3A4 substrate and clarithromycin is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Voriconazole: (Moderate) Consider decreasing the dose of zolpidem if coadministration with voriconazole is necessary. Zolpidem is a CYP3A4 substrate and voriconazole is a strong CYP3A4 inhibitor. Coadministration with strong CYP3A4 inhibitors increased the AUC of zolpidem by 34% to 70%.
Ziconotide: (Moderate) Due to potentially additive effects, dosage adjustments may be necessary if ziconotide is used with a drug that has CNS depressant effects such as anxiolytics, sedatives, and hypnotics. Coadministration of CNS depressants may increase drowsiness, dizziness, and confusion that are associated with ziconotide. Patients taking sedatives with ziconotide may be at higher risk of depressed levels of consciousness. If altered consciousness occurs, consideration of sedative cessation is warranted in addition to ziconotide discontinuation.
Ziprasidone: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of atypical antipsychotics and zolpidem due to the risk for additive CNS depression and next-day psychomotor impairment; dose adjustments may be necessary. Limit the dose of Intermezzo sublingual tablets to 1.75 mg/day.
Zuranolone: (Major) Avoid the use of multiple sedating agents due to the risk for additive CNS depression. If use is necessary, consider a downward dosage adjustment of either or both medications, especially in patients with additional risk factors for sedation-related harm.

Adverse Reaction

bone fractures, confusion, dizziness, drowsiness, fatigue, headache, impaired cognition, intracranial bleeding, lethargy, psychomotor impairment

Most of the adverse effects associated with zolpidem therapy are dose-dependent and CNS-related. The following CNS-related adverse reactions have been reported with various dosage forms of zolpidem in adults and children: dizziness or lightheadedness (1% to 12% in adults; 23.5% in children) and headache (up to 19% in adults; 12.5% in children). In adults, the following CNS-related adverse reactions have been reported: drowsiness (up to 8%), lethargy (3%), fatigue (1% to 3%), and confusion (1.2%). Some data indicate that most cases of confusion (75%) occur in geriatric patients (i.e., 70 years or older), particularly with doses greater than 10 mg/day. Because zolpidem can cause drowsiness and a decreased level of consciousness (impaired cognition, 0.1% to 1%), there is a higher risk of falls, particularly in the elderly, and subsequent severe injuries. Severe injuries associated with zolpidem use have included hip bone fractures and intracranial bleeding or hemorrhage. One clinical trial analysis showed that 93% of those who experienced a fall were at least 70 years of age and 82% of those patients were receiving more than 10 mg/day. During long-term treatment (28 to 35 nights) with zolpidem at doses up to 10 mg, 3% of patients reported a drugged feeling. Plasma levels of zolpidem may be high enough the morning after use to impair activities requiring mental alertness, such as driving, even if the patient feels fully awake. The risk for next-morning impairment is higher inpatients taking extended-release formulations (i.e., Ambien CR). CNS-related adverse reactions reported in placebo-controlled trials of zolpidem extended-release that would also be expected to occur with other dosage forms of zolpidem include: disorientation (3%), psychomotor impairment (2% to 3%), and balance disorder (2%). Patients receiving Intermezzo, a sublingual formulation of zolpidem indicated for middle-of-the night awakenings, are at increased risk of driving impairment and psychomotor impairment if the drug is administered with fewer than 4 hours of bedtime remaining.

amnesia, complex sleep-related behaviors, memory impairment, somnambulism

Sedative-hypnotics can cause complex sleep-related behaviors such as phone calls, sexual activity, preparing and eating food, somnambulism, or driving while not fully awake and in some cases having no memory of the event. These behaviors appear to be more frequent with nonbenzodiazepine benzodiazepine-receptor agonists (NBRAs), such as zolpidem, than other sedative-hypnotics and may occur with or without the concomitant use of alcohol or other CNS depressants. Although rare, serious injuries or death have occurred in patients with and without a history of such behaviors, even at the lowest recommended doses; the behaviors can occur after just one dose. Patients should be informed of the risks before receiving any medication from this class, including instructions to discontinue the medication if they experience a sleep-related episode and to contact their health care provider immediately. Health care professionals

and patients are encouraged to report adverse events or side effects related to the use of NBRAs to the FDA MedWatch Safety Information and Adverse Event Reporting Program. Memory disorders including memory impairment, amnesia, and anterograde amnesia were reported in 1% to 3% of patients in placebo-controlled trials with zolpidem extended-release tablets. No consistent evidence of next-day memory impairment has been demonstrated in controlled studies with the use of zolpidem immediate-release tablets in adults utilizing objective measures of memory. However, in one study of zolpidem 10 to 20 mg/day, there was a significant decrease in next morning recall (i.e., anterograde amnesia) of information presented during the times of peak medication effect (i.e., 90 minutes post-dose). Anterograde amnesia occurs predominantly at doses higher than 10 mg/day. Roughly 1% of patients may experience amnesia of this type at normally prescribed dosages. No patients receiving placebo in clinical trials reported amnesia. Sleep disorder (unspecified) was reported in 1% of zolpidem-treated patients in placebo-controlled trials. Para-suicidal amnestic behavior has been reported rarely.

abnormal dreams, agitation, anxiety, delirium, euphoria, hallucinations, insomnia, nightmares

Anxiety was reported in 2% to 3% of patients in placebo-controlled trials with zolpidem extended-release and in 0.1% to 1% of patients across the entire pre-approval database of zolpidem. Euphoria and insomnia have also been reported in more than 1% of patients on zolpidem therapy. Other paradoxical symptoms of CNS stimulation with zolpidem therapy occur infrequently (i.e., 0.1% to 1%) and include agitation and abnormal dreams (e.g., nightmares). If there is evidence of CNS stimulation, zolpidem should be discontinued. Other thought or behavioral changes include decreased inhibition (e.g., aggressiveness or extroversion), bizarre behavior (unspecified), depersonalization (1%), and auditory/visual hallucinations. Visual hallucinations may also occur when zolpidem is prescribed concurrently with certain antidepressant-type medications. Hallucinations have been reported in clinical trials in less than 1% of adult patients and 7.4% of pediatric patients. Sporadic cases are reported in the literature, with durations of hallucinations lasting from 30 minutes to 7 hours after zolpidem administration. Dosage reduction or discontinuation of treatment should be considered if hallucinations occur. Delirium has also been reported in postmarketing data. The emergence of new behavioral signs or symptoms of concern during zolpidem treatment should receive careful and immediate attention.

depression, suicidal ideation

Depression has been reported in 1% to 2% of zolpidem-treated patients in placebo-controlled trials. Zolpidem, like all central nervous system (CNS) depressants, should be used cautiously in patients with symptoms of depression. These patients may

experience suicidal ideation and be more likely to intentionally overdose on medications. Worsening of depression, including suicidal ideation and suicidality (including completed suicides), has been reported in association with the use of sedative/hypnotics. Zolpidem should be prescribed in the smallest quantity consistent with good patient management to reduce the risk of overdose. Data from a clinical trial in which patients receiving a selective serotonin reuptake inhibitor (SSRI) were given zolpidem revealed that 4 of the 7 discontinuations during double-blind treatment with zolpidem (n = 95) were associated with impaired concentration, continuing or aggravated depression, and manic reaction; one patient treated with placebo (n = 97) was discontinued from treatment after a suicide attempt. Para-suicidal amnestic behavior has also been reported in non-depressed patients who reported concomitant alcohol use at the time of zolpidem administration, which increases the risk for complex sleep-related behaviors or mental status changes.

angina, arrhythmia exacerbation, chest pain (unspecified), edema, hypertension, hypotension, myocardial infarction, orthostatic hypotension, palpitations, phlebitis, pulmonary edema, pulmonary embolism, sinus tachycardia, syncope, ventricular tachycardia

Palpitations have been reported in placebo-controlled clinical trials at a rate of 2% of zolpidem-treated patients compared to no cases in the placebo arm. Other cardiovascular-related adverse events that occurred in 0.1% to 1% of patients include chest pain (unspecified) or chest discomfort, orthostatic hypotension, syncope, edema, cerebrovascular disorder, hypertension, and sinus tachycardia. Angina pectoris, arrhythmia exacerbation, arteritis, circulatory failure, extrasystoles, hypertension aggravated, myocardial infarction, phlebitis, pulmonary embolism, pulmonary edema, varicose veins, hypotension, and ventricular tachycardia have been reported in less than 0.1% of patients.

respiratory depression

During clinical trial evaluation, the 10 mg zolpidem dose did not show respiratory depressant effects in healthy subjects or patients with mild or moderate chronic obstructive pulmonary disease (COPD); however, a reduction in the Total Arousal Index, a reduction in the lowest oxygen saturation, and an increase in the times of oxygen desaturation below 80% to 90% were observed in patients with mild to moderate sleep apnea receiving zolpidem compared to placebo. Respiratory depression occurred in less than 0.1% of patients during premarketing evaluation. Respiratory insufficiency has been reported during postmarketing use, primarily in patients with pre-existing respiratory impairment such as severe COPD or sleep apnea. Concomitant use of CNS depressants, particularly opioids, can also increase the risk for respiratory depression.

anaphylactic shock, anaphylactoid reactions, angioedema, laryngeal edema

Allergy was reported in 4% of patients treated with zolpidem doses of 10 mg/day or less in placebo-controlled trials lasting up to 35 nights. Anaphylactoid reactions (e.g., angioedema) are rare but may occur with sedative-hypnotics, such as zolpidem, and may become evident as early as the initial dose. Airway obstruction resulting from angioedema of the tongue, glottis, or larynx (laryngeal edema) may be fatal. Other symptoms suggestive of anaphylaxis may include shortness of breath and throat closing. Anaphylactic shock and facial edema were reported in less than 0.1% of patients across the entire pre-approval database of zolpidem. Patients should be instructed on the appropriate action in the event of an allergic reaction. Treatment with zolpidem should not be re-initiated in patients who experience angioedema.

acne vulgaris, bullous rash, erythema, flushing, furunculosis, hyperhidrosis, oral ulceration, pallor, photosensitivity, pruritus, rash, urticaria

Dermatologic-related adverse reactions reported during administration of extended-release zolpidem include: rash (unspecified) (1% to 2%), skin wrinkling (1%), and urticaria (0.1% to 1%). Pruritus, increased sweating (hyperhidrosis), and pallor have been reported in 0.1% to 1% of patients taking immediate-release zolpidem. Adverse reactions reported in less than 0.1% of these patients include: acne vulgaris, bullous rash, dermatitis, furunculosis, flushing, and photosensitivity reaction. Localized application site reactions have been reported during postmarketing use of the sublingual tablet for middle-of-the-night administration, including oral ulceration, blisters, and mucosal inflammation. In an open-label study of the effects of chronic daily administration of sublingual zolpidem (n = 60), one patient developed transient sublingual erythema, and one patient developed transient paresthesia of the tongue. Single doses of zolpidem lingual spray showed no sign of oral irritation in studies of healthy volunteers.

abdominal pain, anorexia, constipation, dental caries, diarrhea, dysgeusia, dyspepsia, dysphagia, eructation, flatulence, gastritis, gastroesophageal reflux, GI obstruction, hemorrhoids, hiccups, hypersalivation, nausea, tenesmus, vomiting, weight loss, xerostomia

Many gastrointestinal (GI) adverse events resulting from zolpidem use appear to be dose related. The most commonly observed in placebo-controlled trials with dosages up to 10 mg/day PO were diarrhea (1% to 3%) and constipation (2%). Constipation was reported less frequently (0.1% to 1% of patients) across the entire pre-approval database of

zolpidem. Xerostomia (3%) and appetite disorder (1%) were reported in placebo-controlled trials of zolpidem. GI events reported in more than 1% of subjects across the entire pre-approval database of zolpidem include: dyspepsia, hiccups, and nausea. Nausea was reported more frequently (7%) in a 3-week trial of zolpidem extended-release at a dose of 12.5 mg/day PO. GI events occurring in 0.1% to 1% of patients include: anorexia, dysphagia, dysgeusia, flatulence, gastroenteritis, and vomiting. GI events occurring in less than 0.1% of patients include: weight loss, enteritis, eructation, esophageal spasm, gastritis, hemorrhoids, GI obstruction (i.e., intestinal obstruction), rectal hemorrhage, tenesmus, altered saliva, hypersalivation, and dental caries. Abdominal pain has been reported, but the incidence is similar to that of placebo in clinical trials. Abdominal discomfort or tenderness, frequent bowel movements, and gastroesophageal reflux were reported in 1% of patients in a 3-week trial of zolpidem extended-release at a dose of 12.5 mg/day PO. During a clinical trial of the sublingual zolpidem tablet for middle-of-the night awakenings, nausea occurred at the same rate in treated patients and those receiving placebo (1%); however, GI disorders (unspecified) were reported in 4% of the zolpidem group vs. 2% with placebo. In a separate study using a 1.75 mg dose, similar incidences of adverse reactions were reported.

arthralgia, asthenia, back pain, muscle cramps, myalgia, myasthenia, restless legs syndrome (RLS)

Back pain was reported at an incidence of 3% to 4% in placebo-controlled trials of zolpidem immediate-release and extended-release tablets. Musculoskeletal and connective tissue disorder-related adverse reactions reported in placebo-controlled clinical trials in adults and elderly receiving zolpidem extended-release tablets at a dosage of 6.25 mg/day to 12.5 mg/day include: neck pain (1% to 2%), neck injury (1%), involuntary muscle contractions (1%), contusion (1%), muscle cramps (2%), arthralgia (2%), myalgia (4%), and asthenia (1%). Arthralgia, myalgia, and asthenia were reported in more than 1% of subjects across the entire pre-approval database of zolpidem. Arthritis was reported in 0.1% to 1% of patients across the entire pre-approval database of zolpidem. Musculoskeletal system-related adverse events reported in less than 0.1% of patients across the entire pre-approval database of zolpidem include: arthrosis, muscle weakness (myasthenia), sciatica, pain, falls, restless legs syndrome (RLS) and tendinitis.

chills, fever, hot flashes, infection, influenza, malaise, pharyngitis, sinusitis

Influenza-like symptoms (1% to 3%), sinusitis (4%), pharyngitis (3%), nasopharyngitis (6%), throat irritation (1%), dry throat (1%), and lower respiratory and upper respiratory tract infection (1%) were reported in placebo-controlled trials of zolpidem. Infection,

fever, and malaise were reported across the entire pre-approval database in 0.1% to 1% of patients. Chills (rigors) and hot flashes occurred in less than 0.1% of patients.

bronchospasm, cough, dyspnea, epistaxis, hypoxia, laryngitis, rhinitis

Respiratory-related adverse reactions reported in 0.1% to 1% of patients with zolpidem administration include: bronchitis, cough, dyspnea, and rhinitis. Those occurring in less than 0.1% patients include: bronchospasm, epistaxis, hypoxia, laryngitis, and pneumonia.

tinnitus

Labyrinthitis and otitis externa were reported in 1% of zolpidem-treated patients in placebo-controlled trials of zolpidem extended-release at doses of 6.25 mg/day to 12.5 mg/day. Across the entire pre-approval database of zolpidem immediate-release tablets, otitis externa and otitis media were reported in less than 0.1% of patients. Tinnitus has been reported in 0.1% to 1% of patients.

blurred vision, conjunctival hyperemia, conjunctivitis, diplopia, lacrimation, ocular irritation, ocular pain, photopsia, visual impairment

Ophthalmic-related adverse reactions reported with zolpidem administration include: visual impairment (up to 3%), conjunctival hyperemia (2%), blurred vision (2%), altered visual depth perception (1%), and asthenopia (1%). Diplopia and abnormal vision have been reported in more than 1% of subjects. Adverse reactions reported in 0.1% to less than 1% of patients include: ocular irritation, ocular pain, and scleritis. Conjunctivitis, corneal ulceration, abnormal lacrimation, abnormal accommodation, glaucoma, and photopsia have occurred in less than 0.1% of patients.

mastalgia, menorrhagia, vaginal irritation, vaginitis

Menorrhagia and vulvovaginal dryness (vaginal irritation) were reported in 1% of extended-release zolpidem-treated female patients in placebo-controlled trials. Menstrual disorder and vaginitis have been reported in 0.1% to 1% of patients taking immediate-release zolpidem. Breast fibroadenosis, breast neoplasm, and breast pain (mastalgia) have also occurred in less than 0.1% of patients.

cystitis, dysuria, impotence (erectile dysfunction), increased urinary frequency, nocturia, polyuria, renal failure (unspecified), urinary incontinence, urinary retention

Dysuria was reported in 1% of patients receiving zolpidem extended-release in clinical trials and in less than 0.1% of patients in all pre-marketing studies of zolpidem. Urinary tract infection was also reported, affecting 0.1% to 1% of patients on immediate-release zolpidem and more than 1% of patients on an extended-release formulation. Cystitis and urinary incontinence were less common, affecting 0.1% to 1% of patients on all formulations. Rare reactions (occurring in less than 0.1% of patients on any formulation) include impotence (erectile dysfunction), acute renal failure (unspecified), increased urinary frequency, nocturia, polyuria, pyelonephritis, renal pain, and urinary retention.

azotemia, gout, hypercholesterolemia, hyperglycemia, hyperlipidemia

Hyperglycemia and thirst were reported in 0.1% to 1% of patients across the entire pre-approval database of zolpidem. Adverse reactions reported in less than 0.1% of patients include: gout, hypercholesterolemia, hyperlipidemia, increased BUN (azotemia), and periorbital edema.

elevated hepatic enzymes, hepatic encephalopathy, hyperbilirubinemia, jaundice

Elevated hepatic enzymes were reported across the entire pre-approval database of zolpidem. During clinical trial evaluation, increased SGPT occurred in 0.1% to 1% of patients; increases in alkaline phosphatase and SGOT occurred in less than 0.1% of patients. Other hepatic and biliary system-related adverse events included: abnormal hepatic function (0.1% to 1% of patients) and hyperbilirubinemia (less than 0.1% of patients). Gamma-aminobutyric acid (GABA) agonists, such as zolpidem, have been associated with the development of hepatic encephalopathy in patients with hepatic insufficiency. Acute hepatocellular, and cholestatic or mixed liver injury with or without jaundice (i.e., bilirubin greater than 2 times the upper limit of normal (ULN), alkaline phosphatase at least 2 times the ULN, or transaminases at least 5 times the ULN) have been reported during postmarketing use of zolpidem.

anemia, leukopenia, lymphadenopathy, purpura, thrombosis

Anemia, macrocytic anemia, hyperhemoglobinemia, leukopenia, lymphadenopathy, purpura, increased erythrocyte sedimentation rate (ESR), and thrombosis were reported in less than 0.1% of patients across the entire pre-approval database of zolpidem.

appetite stimulation, ataxia, dysarthria, emotional lability, hypoesthesia, libido decrease, migraine, paresthesias, tetany, tremor, vertigo, yawning

Apathy (1%) and hypoesthesia (2%) were reported in placebo-controlled trials of zolpidem extended-release tablets. Across the entire pre-approval database of zolpidem, apathy occurred in less than 0.1% of patients and hypoesthesia occurred in 0.1% to 1% of patients. Other central and peripheral nervous system-related adverse events reported across the entire pre-approval database of zolpidem occurring in greater than 1% of patients include: ataxia and vertigo. Those reported infrequently (occurring in 0.1% to 1% of patients) include: difficulty concentrating, dysarthria, emotional lability or mood swings, illusion, leg cramps, migraine, nervousness, paresthesias, sleeping (after daytime dosing), speech disorder, stupor, tremor. Rare reaction (reported in less than 0.1% of patients) include: abnormal gait, abnormal thinking, aggressive reaction, appetite stimulation, libido decrease, delusion, dementia, depersonalization, dysphasia, feeling strange, hypokinesia, hypotonia, hysteria, intoxicated feeling, manic reaction, neuralgia, neuritis, neuropathy, neurosis, panic attacks, paresis, personality disorder, tetany, and yawning.

tolerance

Tolerance has been reported in less than 0.1% of patients across the entire pre-approval database of zolpidem.

physiological dependence, psychological dependence, withdrawal

Drug dependency (psychological dependence or physiological dependence) to zolpidem may be related to dosage and duration of treatment. There have been postmarketing reports of abuse, dependence and withdrawal with zolpidem. A dosage-tapering schedule may be adopted if drug dependency is suspected. Withdrawal of some hypnotics can precipitate a rebound insomnia and older adults may be more susceptible to sleep impairment following discontinuation of therapy. Rebound insomnia occurs rarely following discontinuation of zolpidem after continuous administration for 28 days or less. Sedative/hypnotics have produced withdrawal signs and symptoms following abrupt discontinuation. These reported symptoms range from mild dysphoria and insomnia to a withdrawal syndrome that may include abdominal and muscle cramps, vomiting, sweating, tremors, convulsions, and delirium. The following adverse events, which are considered to meet the DSM-III-R criteria for uncomplicated sedative/hypnotic withdrawal, were reported within 48 hours following the last zolpidem immediate-release or ER dosage: fatigue, nausea, flushing, lightheadedness, uncontrolled crying, emesis, stomach cramps, panic attack, nervousness, and abdominal discomfort. These reported adverse events occurred at an incidence of 1% or less. However, available withdrawal data cannot provide a reliable estimate of the incidence.

Description

Zolpidem is a non-benzodiazepine sedative-hypnotic used for the treatment of insomnia in adults. Zolpidem is available in various orally administered formulations (e.g., Ambien, Ambien CR) for difficulty with sleep initiation; one formulation (Intermezzo) was granted FDA approval for difficulty falling back to sleep following middle-of-the-night awakenings. Zolpidem products for sleep initiation have a rapid onset of action and should be taken immediately before retiring. When used for middle-of-the-night awakenings, there should be at least 4 hours of bedtime remaining before the planned time of waking. Zolpidem can cause next-day psychomotor impairment. Unlike benzodiazepines, there is a lack of myorelaxant, anxiolytic, and anticonvulsant effects with zolpidem at therapeutic doses. Some sedative-hypnotics, including zolpidem, have been associated with complex sleep-related behaviors (e.g., making phone calls, engaging in sexual activity, eating, or driving while not fully awake). All sedative-hypnotics have the potential to cause sleep-related behaviors; therefore, patients should be informed of the risks prior to receiving any medication from this class. While the drug has been studied in pediatric patients (6 to 17 years of age), zolpidem was not shown effective at treating pediatric insomnia and adverse events were frequent.

Mechanism Of Action

Zolpidem is a gamma-aminobutyric acid-A (GABA-A) receptor positive modulator thought to exert its therapeutic effect in the treatment of insomnia through binding at the omega-1 subunit of the GABA-A receptor site. Binding at the omega-1 subunit increases the frequency of chloride channel opening, which results in the inhibition of neuronal excitation. Zolpidem is chemically unrelated to benzodiazepines, which non-selectively interact with omega-receptor subtypes at the GABA-A receptor site. Zolpidem binds to the omega-1 subunit with greater affinity than the omega-2 or omega-3 subunits and has no appreciable binding to omega-5, which may explain its lack of myorelaxant, anxiolytic, and anticonvulsant effects compared to benzodiazepines. Sleep studies in humans indicate that zolpidem normally preserves deep sleep (stages 3 and 4) and that changes in rapid eye movement (REM) sleep are minor and inconsistent. Anterograde amnesia appears to be more likely with dosages higher than 10 mg. In one study of zolpidem 10 mg or 20 mg, a significant decrease in next morning recall of information was observed during the times of peak medication effect (i.e., 90 minutes post-dose). Signs and symptoms of withdrawal may occur with rapid dose reduction or abrupt discontinuation. Zolpidem has no appreciable binding affinity for dopaminergic (D2), serotonergic (5-HT2), adrenergic, histaminic, or muscarinic receptors.

Pharmacokinetics

Zolpidem is administered orally as an immediate-release tablet, immediate-release capsule, controlled-release tablet, and sublingual tablet. It is about 92% bound to plasma protein. Hepatic metabolism through CYP isoenzymes produces inactive metabolites that are primarily excreted in the urine. Immediate-release zolpidem has not been shown to accumulate in young adults with administration up to 2 weeks. The mean elimination half-life of the immediate-release tablets is 2.6 hours (range 1.4 to 4.5 hours) for the 5-mg dose and 2.5 hours (ran: 1.4 to 3.8 hours) for the 10-mg tablets in patients with normal hepatic and renal function. Mean elimination half-life of the 7.5-mg dose of zolpidem immediate-release capsules is 3.7 hours (range: 1.2 to 8.3 hours) under fasting conditions. The half-life and plasma protein binding are similar between the extended-release and immediate-release tablet formulations. The mean elimination half-life of the Edluar sublingual tablet is 2.85 hours (range: 1.57 to 6.73 hours) and 2.65 hours (range: 1.75 to 3.77 hours) after a single dose of 5 or 10 mg, respectively, in healthy adult volunteers. The elimination half-life of a single 3.5 mg dose of the Intermezzo sublingual tablet for middle-of-the-night awakenings is about 2.5 hours (range: 1.4 to 3.6 hours). Blood concentrations of zolpidem above 50 ng/mL appear to be capable of impairing driving to a level that increases the risk of a motor vehicle accident. Data show that the risk for next-morning impairment may be higher for the extended-release than the immediate-release formulation.

Affected cytochrome P450 (CYP450) isoenzymes and drug transporters: CYP3A4, CYP2C9, CYP2D6, CYP2C19
Based on in vitro data, approximately 61% of zolpidem is metabolized by CYP3A4, with minor metabolism pathways by CYP2C9 (22%), CYP1A2 (14%), CYP2D6 (less than 3%), and CYP2C19 (less than 3%). Consider a lower dose of zolpidem when a potent CYP3A4 inhibitor and zolpidem are given together. Use of a potent CYP3A4 inducer in combination with zolpidem may decrease the efficacy of zolpidem and is not recommended.

Route-Specific Pharmacokinetics

- **Oral Route**

Immediate-release tablets (e.g., Ambien): Zolpidem is rapidly absorbed after oral administration. Peak medication effects of the immediate-release tablet occur within 90 minutes of a single oral dosage. In single-dose studies in subjects administered 5 mg and 10 mg zolpidem, the mean peak concentrations (Cmax) were 59 (range: 29 to 113) and 121 (range: 58 to 272) ng/mL, respectively, occurring at a mean time (Tmax) of 1.6 hours for both strengths. The presence of food reduces the amount of absorption and

increases the time taken to achieve maximum concentration, delaying sleep onset. Therefore, all formulations of zolpidem should be taken on an empty stomach versus after a meal.

Immediate-release capsules: Following a single dose administration of 7.5 mg zolpidem capsules and 10 mg zolpidem tablets to healthy volunteers in fasting conditions, the Cmax and AUC of zolpidem capsules and tablets were similar. Mean zolpidem Cmax and AUC decreased by 39% and 8%, respectively, with a high-fat, high-calorie meal compared to fasted conditions following administration of zolpidem tartrate capsules in healthy volunteers. The median time to reach zolpidem Tmax was prolonged from 1 hour to 2.8 hours post dose in the presence of food. Therefore, zolpidem capsules should not be taken with or immediately after a meal.

Extended-release (biphasic) tablets (e.g., Ambien CR): Extended-release zolpidem exhibits biphasic absorption characteristics through a coated 2-layer tablet with rapid initial absorption from the GI tract through the first tablet layer similar to zolpidem immediate-release, followed by extended plasma concentrations beyond 3 hours after administration by release of drug through the second layer. Similar to the immediate-release dosage form, zolpidem extended-release should be taken on an empty stomach for faster absorption and sleep-onset. Data show that the risk for next-morning impairment is higher for the extended-release than the immediate-release formulation.

Sublingual Route

Sublingual tablets (e.g., Edluar): Zolpidem sublingual tablets are bioequivalent to the immediate-release tablets with regard to Cmax and AUC. The mean peak plasma concentration occurs at a median time of 82 minutes (range: 30 to 180 minutes) after administration. When the drug is administered within 20 minutes after a high fat meal, the mean AUC and Cmax are decreased by 20% and 31%, respectively, compared to administration while fasting. The median Tmax is prolonged by 28%. Therefore, for faster sleep onset, zolpidem sublingual tablets should not be taken with or immediately after a meal.

Sublingual tablets for middle-of-the-night awakenings (e.g., Intermezzo): These sublingual tablets disintegrate in the sublingual cavity after administration. The mean Tmax is about 35 to 75 minutes. The average Cmax and AUC are higher in women than men. Administration with food decreases the overall Cmax and AUC by 42% and 19% respectively, and increases the Tmax to nearly 3 hours. Therefore, administration with or immediately after a meal is not recommended.

- **Hepatic Impairment**

The pharmacokinetic parameters of zolpidem are affected by hepatic function. In patients with chronic cirrhosis, the immediate-release oral zolpidem Cmax and AUC were 2 and 5 times higher, respectively, than in normal controls and the half-life was also significantly increased to an average of 9.9 hours (range 4.1 to 25.8 hours),

necessitating dosage reductions in the hepatically impaired. Dosage adjustments are also recommended for other zolpidem dosage forms depending on the degree of hepatic impairment.

- **Renal Impairment**

The pharmacokinetic parameters of zolpidem are not appreciably altered in patients with renal impairment. Zolpidem is not removed by hemodialysis; however, the drug does not appear to accumulate in patients with end-stage renal disease with short-term durations (i.e., less than 2 to 3 weeks) of administration.

- **Geriatric**

The pharmacokinetic parameters of zolpidem (both immediate- and extended-release) are altered with advanced age. The mean half-life of both products increased to roughly 3 hours in elderly patients. In comparative trials between younger and older adult subjects receiving zolpidem immediate-release tablets, significant elevations in both Cmax and AUC have been noted. In a pharmacokinetic study of 1.75 mg and 3.5 mg doses of sublingual zolpidem (Intermezzo), the plasma Cmax and AUC following the 3.5 mg dose were 34% and 30% higher, respectively, in elderly subjects compared to non-elderly subjects. The Cmax and AUC following the 1.75 mg dose in elderly subjects were higher than in non-elderly subjects receiving the same dose but lower than the Cmax and AUC in non-elderly subjects receiving the 3.5 mg dose. Zolpidem dosage reductions are recommended for the various dosage forms in the geriatric adult.

- **Gender Differences**

Zolpidem blood concentrations of greater than 50 ng/mL appear to be capable of impairing driving to a level that increases the risk of a motor vehicle accident. Because adult females eliminate zolpidem at a slower rate than adult males, lower initial dosages of zolpidem are recommended in adult females. In simulation and laboratory studies, blood levels more than 50 ng/mL were observed in approximately 15% of women and 3% of men 8 hours after a 10 mg dose of immediate-release zolpidem. Three measurements in women and one in men were 90 ng/mL or higher. Data show that the risk for next-morning impairment is highest for both male and female patients taking the extended-release formulation (i.e., Ambien CR). In pharmacokinetic trials assessing the 12.5 mg dose of extended-release zolpidem, about 33% of women and 25% of men had blood levels more than 50 ng/mL approximately 8 hours after a dose. About 5% of patients had blood levels of 100 ng/mL or higher. In studies of the 6.25 mg extended-release dose, about 15% of adult women and 5% of adult men had blood levels more than 50 ng/mL compared to about 10% of elderly men and women. The Cmax and AUC of sublingually administered zolpidem (Intermezzo) were about 45% higher in adult females compared to males receiving the same dose. In addition, women cleared Intermezzo from the body at a lower rate than men.

Administration

For storage information, see the specific product information within the How Supplied section.

Oral Administration

Oral Solid Formulations

Immediate-release tablet (e.g., Ambien):
Administer immediately before retiring.
Instruct patients to take on an empty stomach to facilitate the onset of sleep. For optimal effect, do not administer with or immediately after a meal.

Immediate-release capsule:
Administer immediately before retiring.
Administer whole; do not open, cut, crush, or chew.
Instruct patients to take on an empty stomach to facilitate the onset of sleep. For optimal effect, do not administer with or immediately after a meal.

Extended-release tablet (e.g., Ambien CR):
Administer immediately before retiring.
Administer the extended-release tablet whole; do not cut, crush, or chew.
Instruct patients to take on an empty stomach to facilitate the onset of sleep. For optimal effect, do not administer with or immediately after a meal.

Other Oral Formulations

Sublingual Administration
Sublingual tablet (e.g., Edluar):
Administer immediately before retiring.
Place the sublingual tablet under the tongue where it can disintegrate; it should not be swallowed whole. Do not administer with water.
Instruct patients to take on an empty stomach to facilitate the onset of sleep. For optimal effect, do not administer with or immediately after a meal.

Sublingual tablet used for middle-of the-night awakening (e.g., Intermezzo):
This product should be used for middle-of-the night awakenings only when there is at least 4 hours of bedtime left before the planned time of waking. Administer only if there are at least 4 hours of bedtime remaining before the planned time of waking.
A Dosing Time Chart and a Dosing Time Tool are provided with the product. Refer to the manufacturer provided "Instructions for Use".

Remove the foil blister containing the tablet from exterior pouch just prior to dosing. The manufacturer recommends leaving the empty pouch where it can be seen throughout the night as a reminder that a dose has been taken.
Place the sublingual tablet under the tongue where it can disintegrate; it should not be swallowed whole.

Maximum Dosage Limits

- **Adults**
  Immediate-release tablets (e.g., Ambien): 10 mg/night PO.
  Immediate-release capsules: 7.5 mg/night PO.
  Extended-release tablets (e.g., Ambien CR): 12.5 mg/night PO.
  Sublingual tablets (e.g., Edluar): 10 mg/night sublingually.
  Sublingual tablets for middle-of the night awakening (e.g., Intermezzo): 1.75 mg sublingually once per night in women and 3.5 mg sublingually once per night in men.
- **Geriatric**
  Immediate-release tablets (e.g., Ambien): The recommended geriatric dose is 5 mg/night PO.
  Extended-release tablets (e.g., Ambien CR): The recommended geriatric dose is 6.25 mg/night PO.
  Sublingual tablets (e.g., Edluar): The recommended geriatric dose is 5 mg/night sublingually.
  Sublingual tablets for middle-of the night awakening (e.g., Intermezzo): The recommended geriatric dose is 1.75 mg sublingually once per night if needed.
- **Adolescents**
  Safety and efficacy have not been established.
- **Children**
  Safety and efficacy have not been established.
- **Infants**
  Safety and efficacy have not been established.
- **Neonates**
  Safety and efficacy have not been established.

Dosage Forms

- Ambien 10mg Tablet
- Ambien 10mg Tablet

- Ambien 10mg Tablet
- Ambien 5mg Tablet
- Ambien 5mg Tablet
- Ambien 5mg Tablet
- Ambien CR 12.5mg Extended-Release Tablet
- Ambien CR 6.25mg Extended-Release Tablet
- Ambien CR 6.25mg Extended-Release Tablet
- Edluar 10mg Sublingual Tablet
- Edluar 10mg Sublingual Tablet
- Edluar 5mg Sublingual Tablet
- Zolpidem Tartrate 1.75mg Sublingual tablet
- Zolpidem Tartrate 10mg Oral tablet
- Zolpidem Tartrate 12.5mg Oral tablet, biphasic release
- Zolpidem Tartrate 3.5mg Sublingual tablet
- Zolpidem Tartrate 5mg Oral tablet
- Zolpidem Tartrate 6.25mg Oral tablet, biphasic release
- Zolpidem Tartrate 7.5mg Oral capsule

Dosage Adjustment Guidelines

Hepatic Impairment

Immediate-release tablets (e.g., Ambien):
-mild to moderate hepatic impairment: 5 mg/night PO if needed.
-severe hepatic impairment: Avoid use as it may contribute to encephalopathy.

Immediate-release capsules:
-mild to moderate hepatic impairment: Avoid use, as dose adjustments are not possible with available strengths.
-severe hepatic impairment: Avoid use as it may contribute to hepatic encephalopathy.

Extended-release tablets (e.g., Ambien CR):
-mild to moderate hepatic impairment: 6.25 mg/night PO if needed.
-severe hepatic impairment: Avoid use as it may contribute to encephalopathy.

Sublingual tablets (e.g., Edluar):
-mild to moderate hepatic impairment: 5 mg/night sublingual if needed.
-severe hepatic impairment: Avoid use as it may contribute to encephalopathy.

Sublingual tablets for middle-of-the night awakening (e.g., Intermezzo):

-mild to moderate hepatic impairment: 1.75 mg sublingual once per night as needed.
-severe hepatic impairment: Avoid use as it may contribute to encephalopathy.

Renal Impairment

It appears that no dosage adjustments are needed. Evidence to date does not indicate a need to use lower doses of zolpidem; however, as a general precaution patients with renal impairment should be closely monitored.

Intermittent hemodialysis
Zolpidem is not removed by hemodialysis. Based on limited study in end-stage renal disease patients undergoing hemodialysis 3 times each week, it appears that no dosage accumulation occurs after 14 or 21 days of daily zolpidem administration. However, close clinical monitoring is recommended as a general precaution.

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